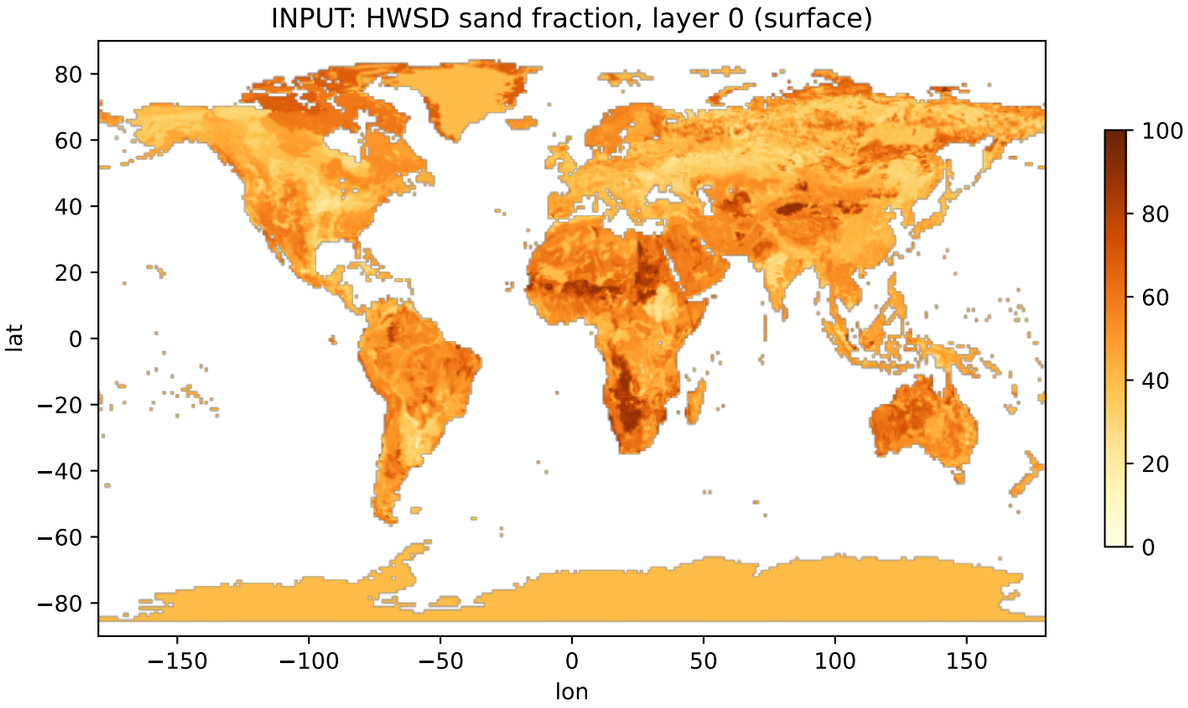
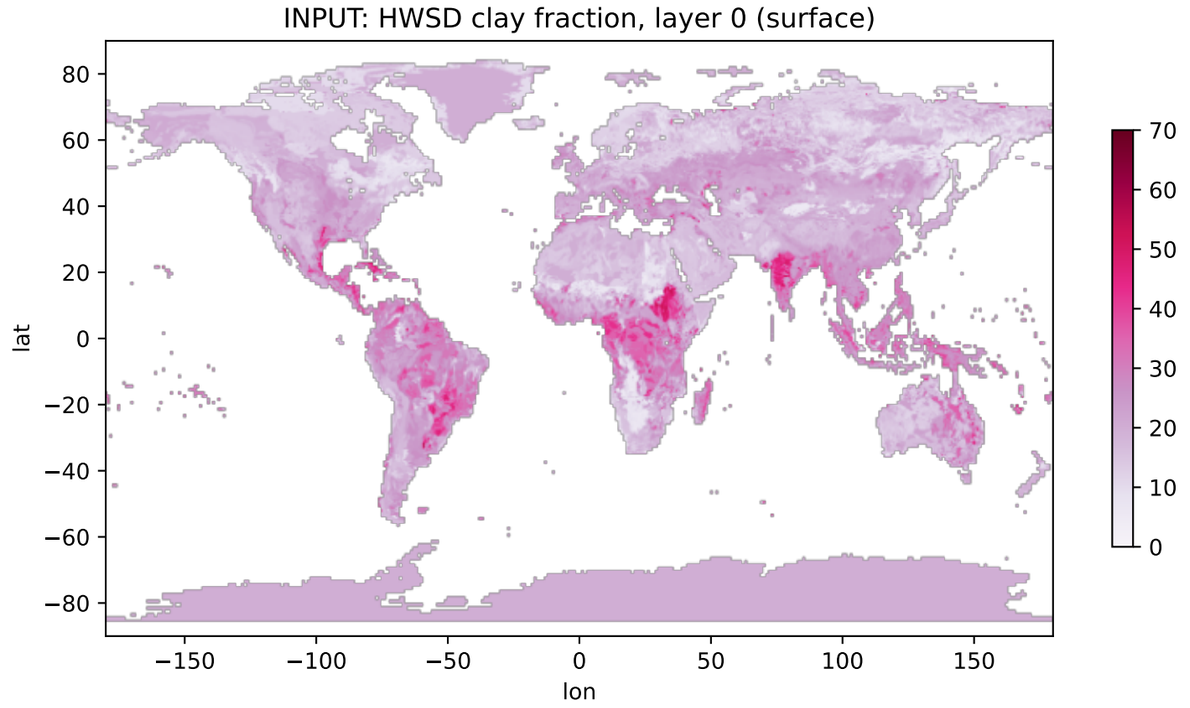
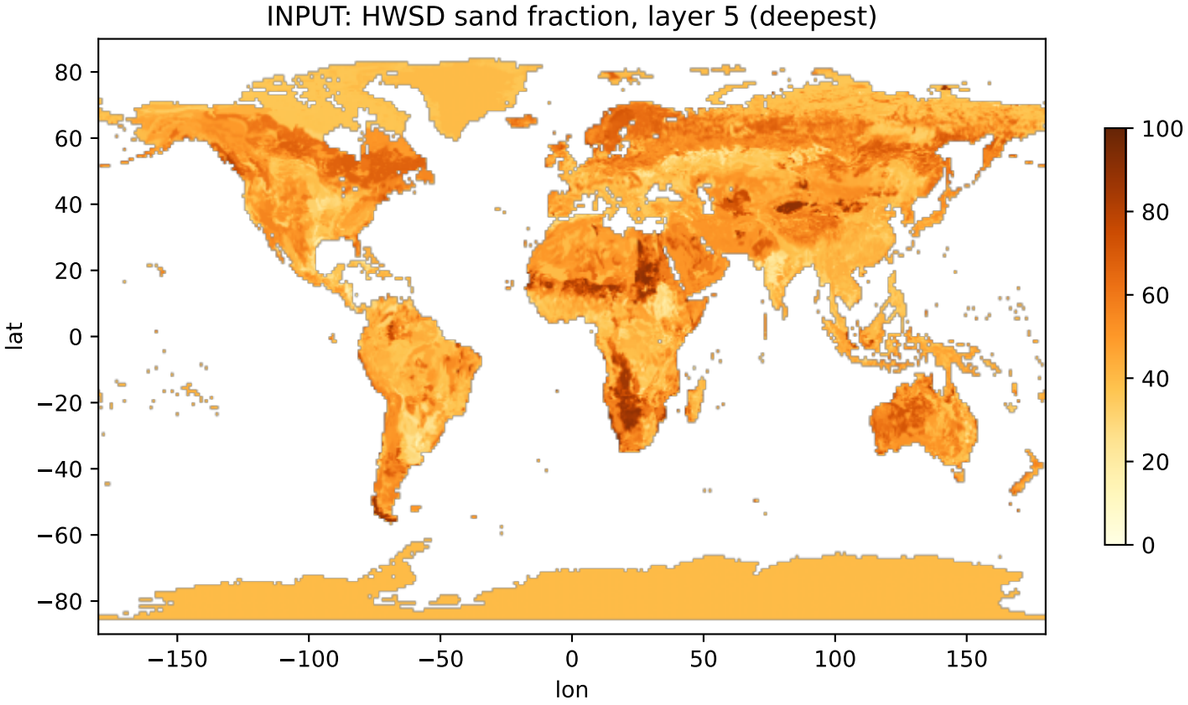
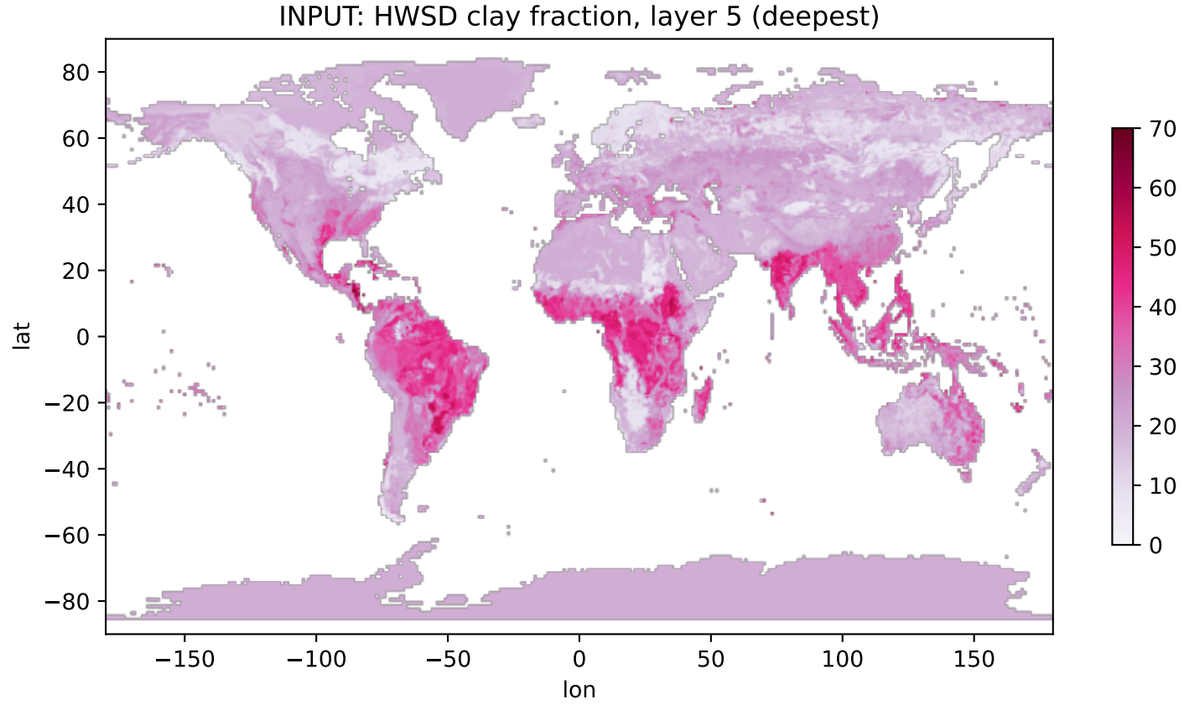
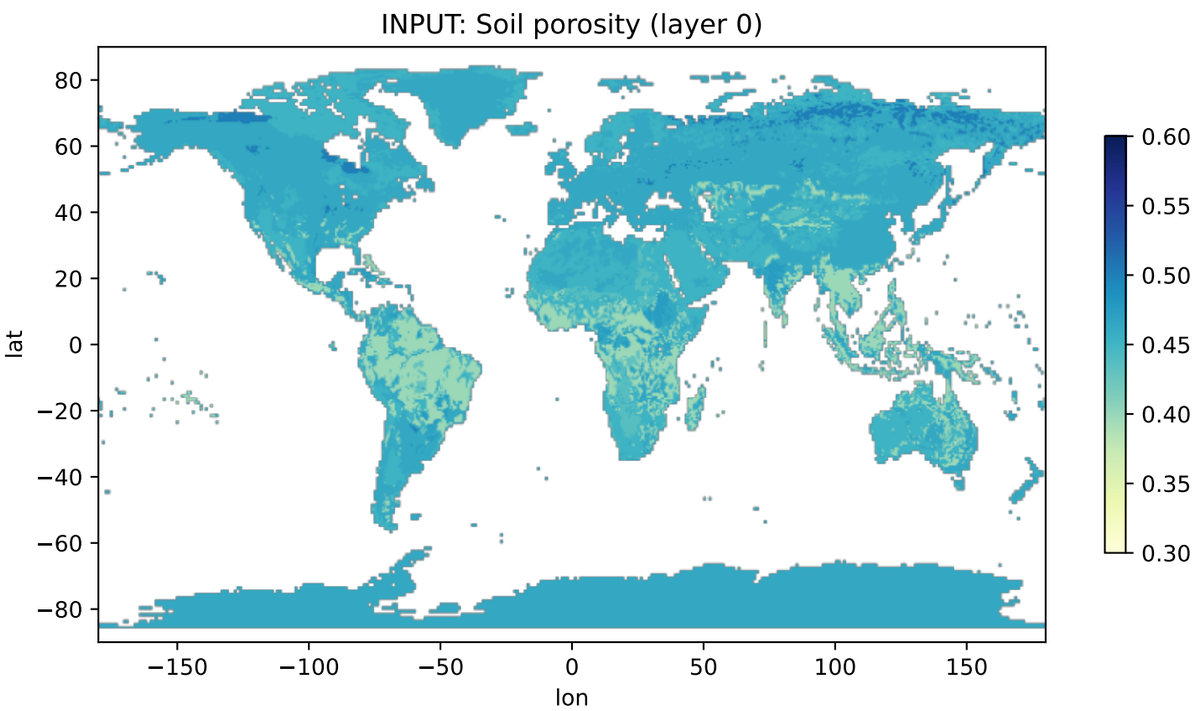
INPUTS: HWSD-derived soil hydraulic & textural properties

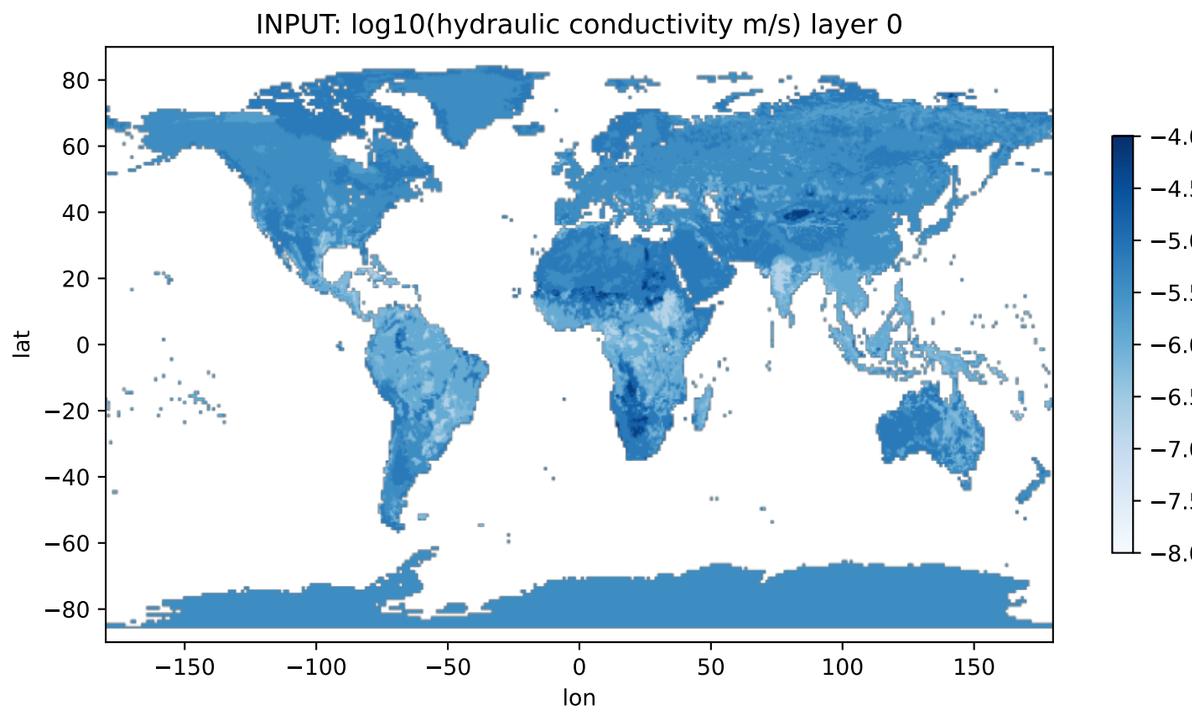




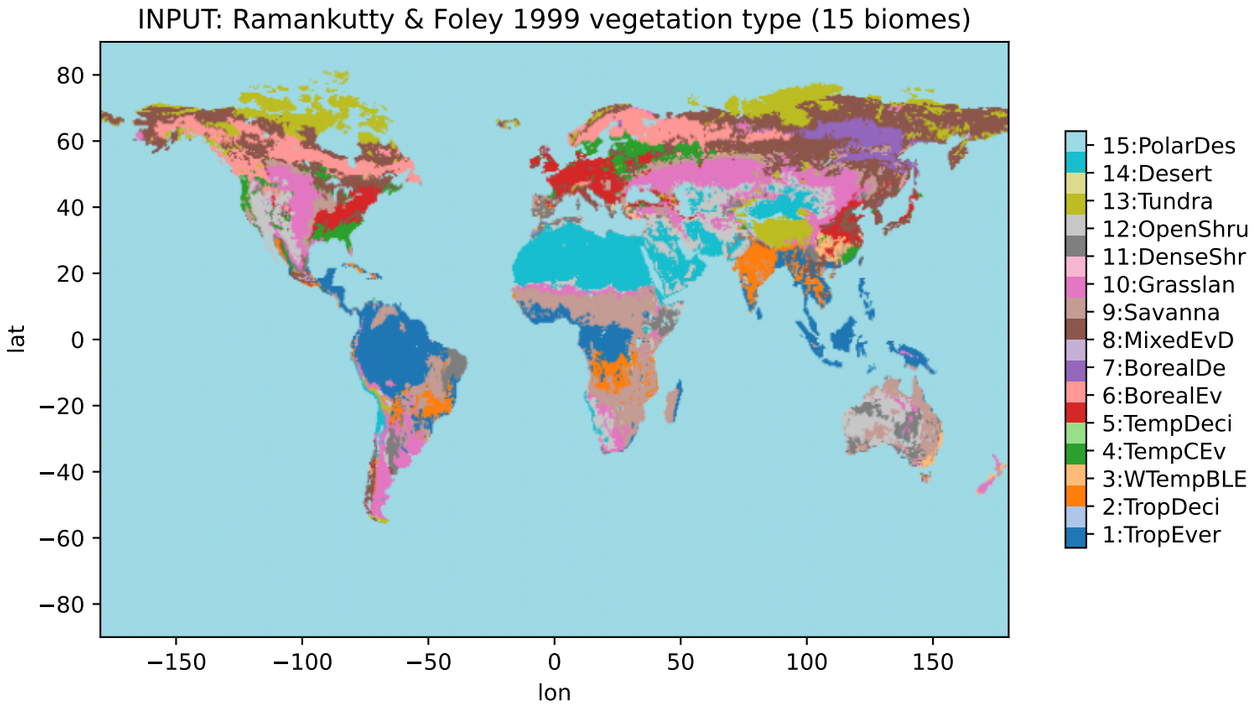


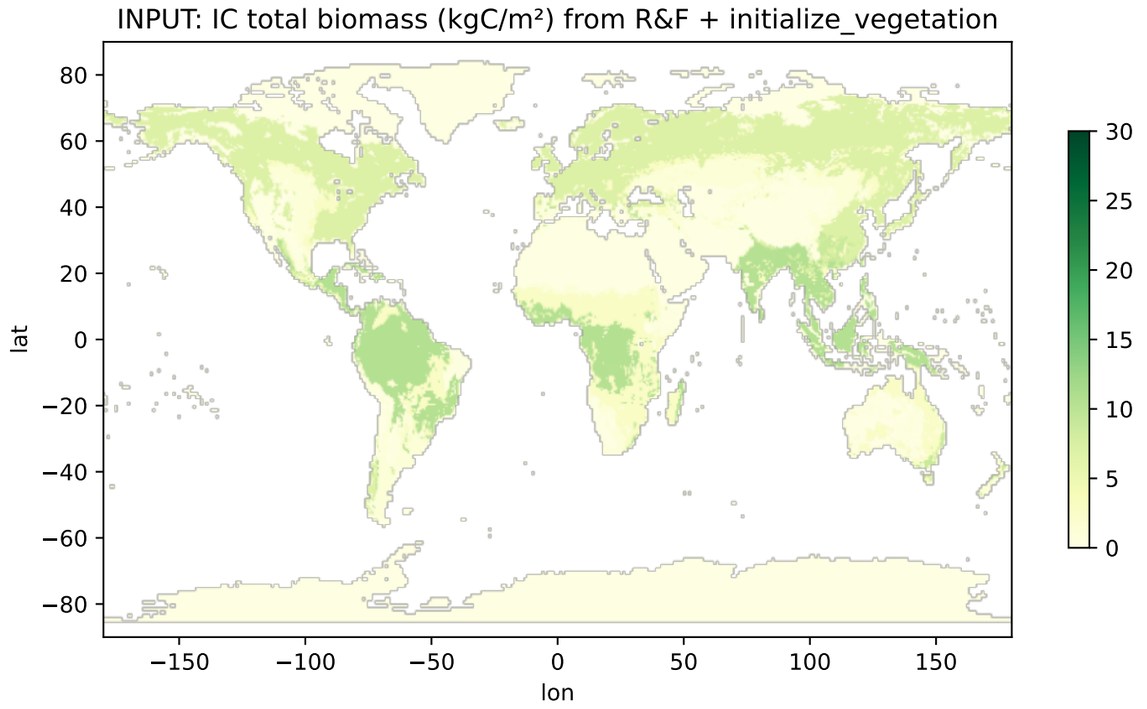


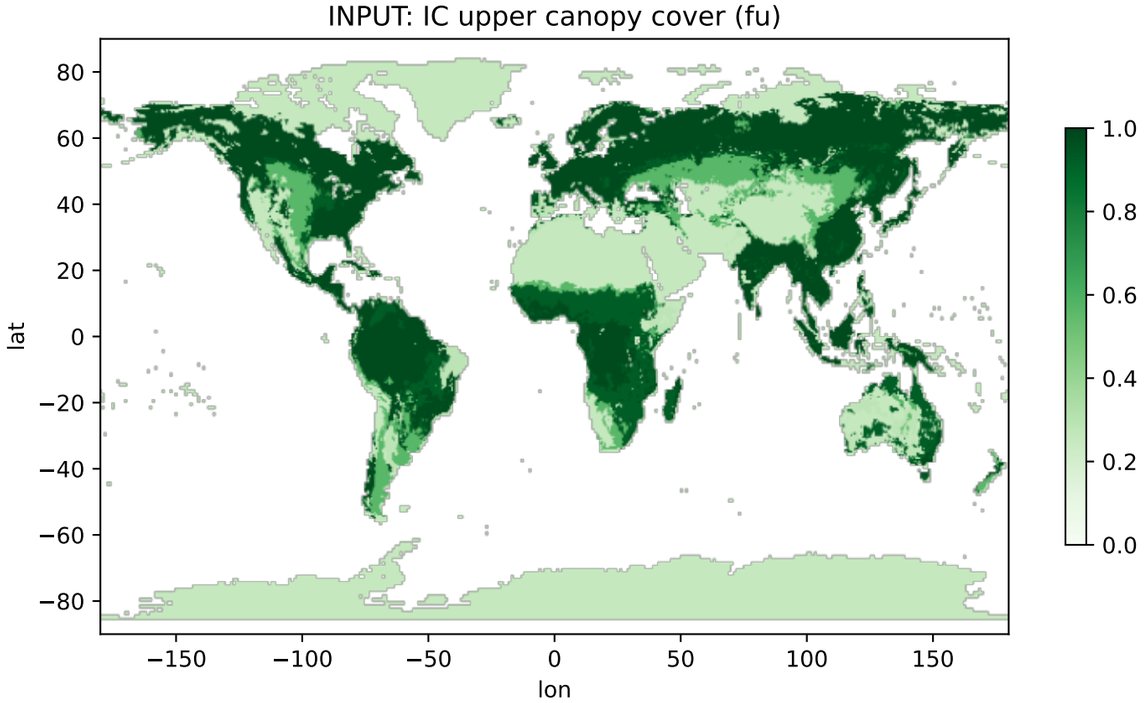


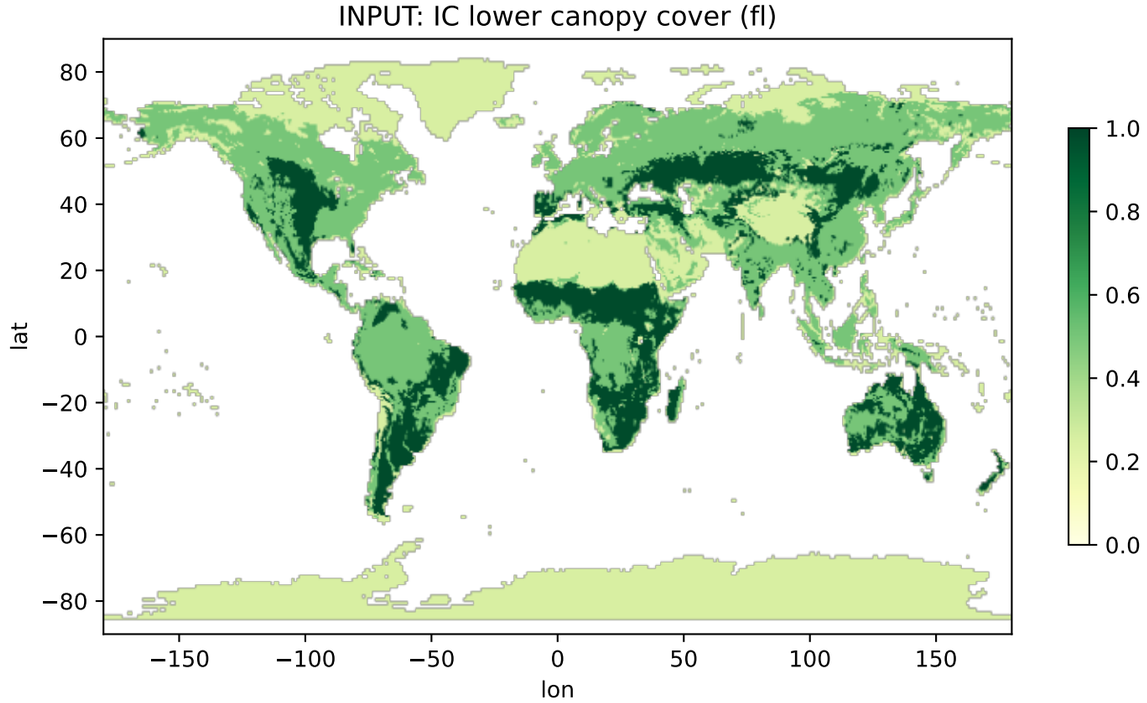


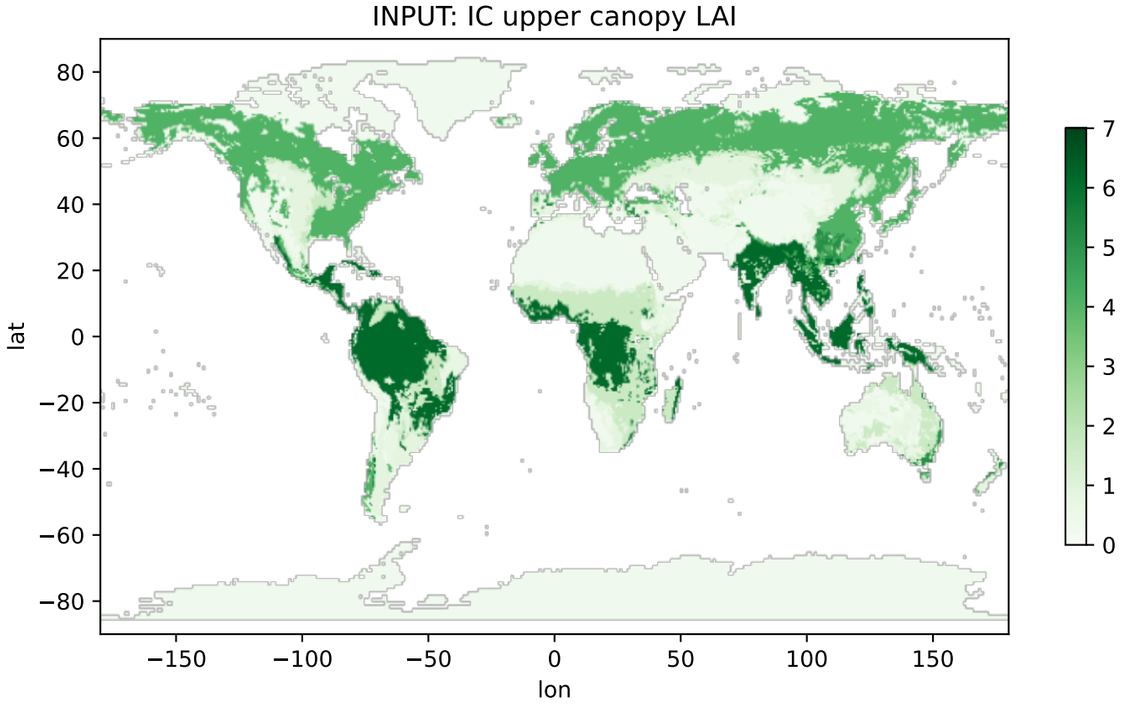
INPUTS: Ramankutty & Foley 1999 vegetation + derived IC

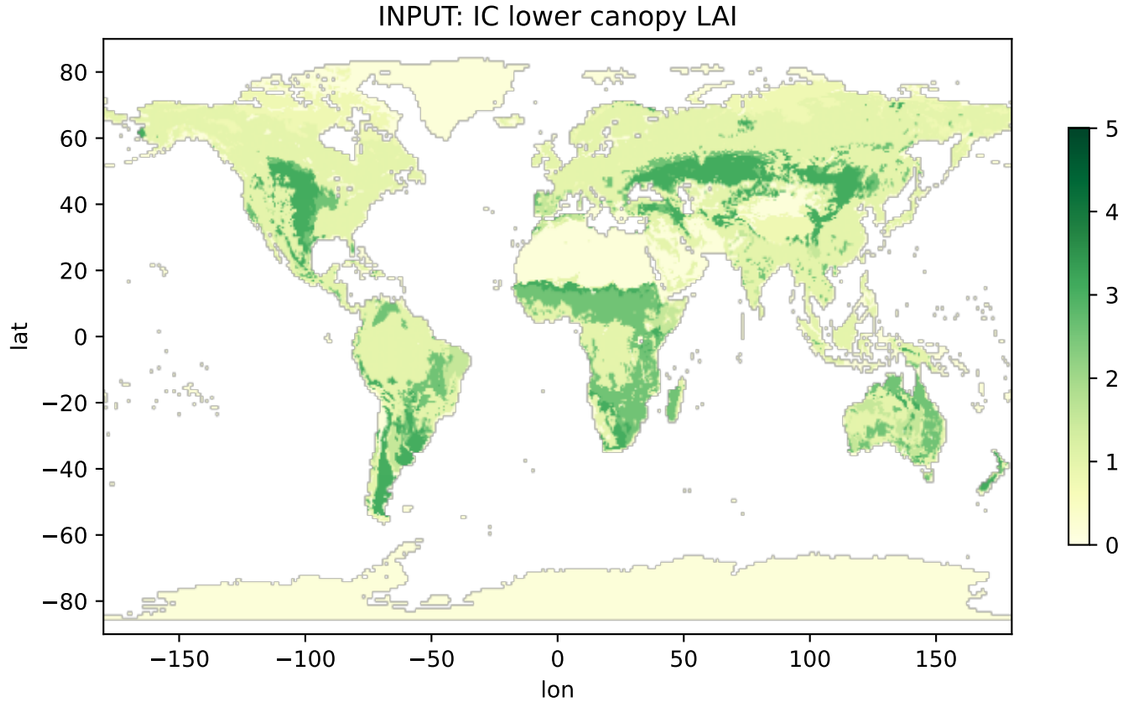




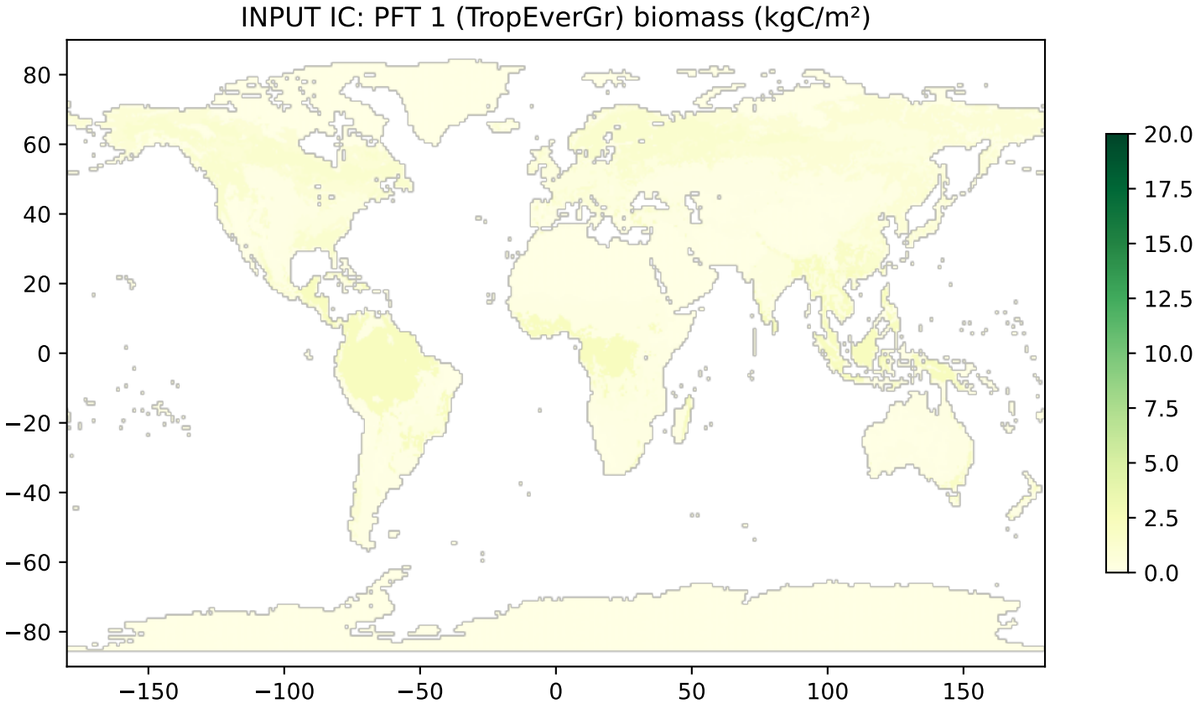


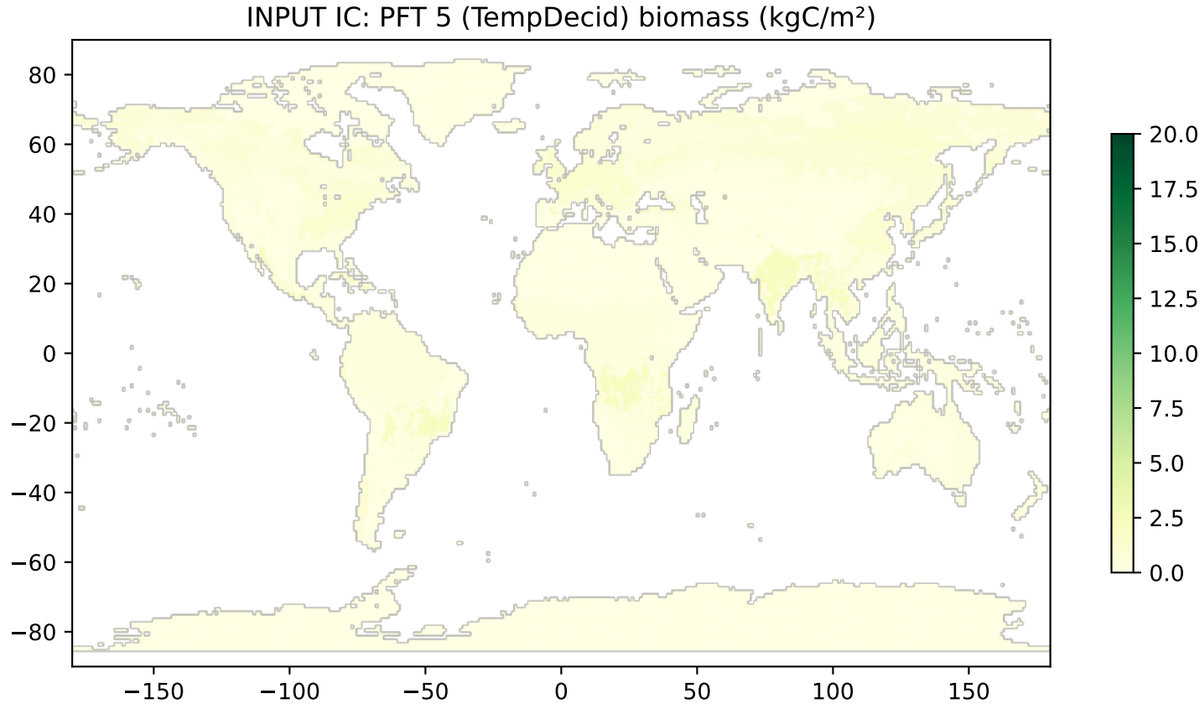


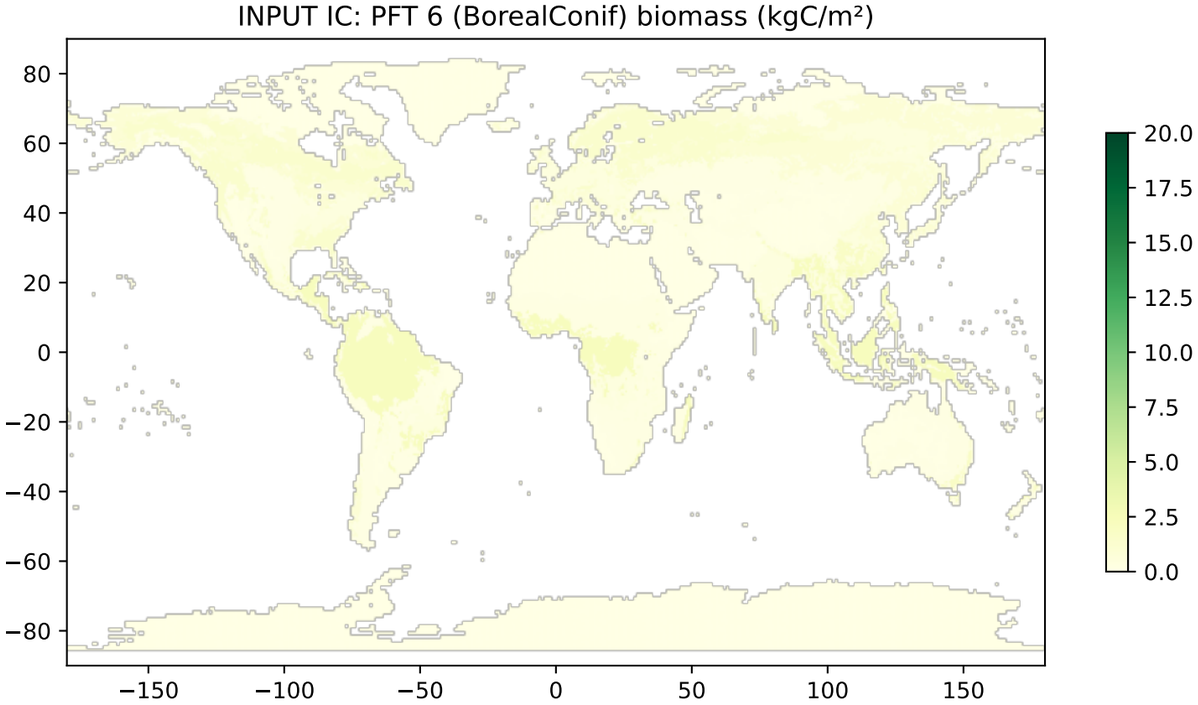


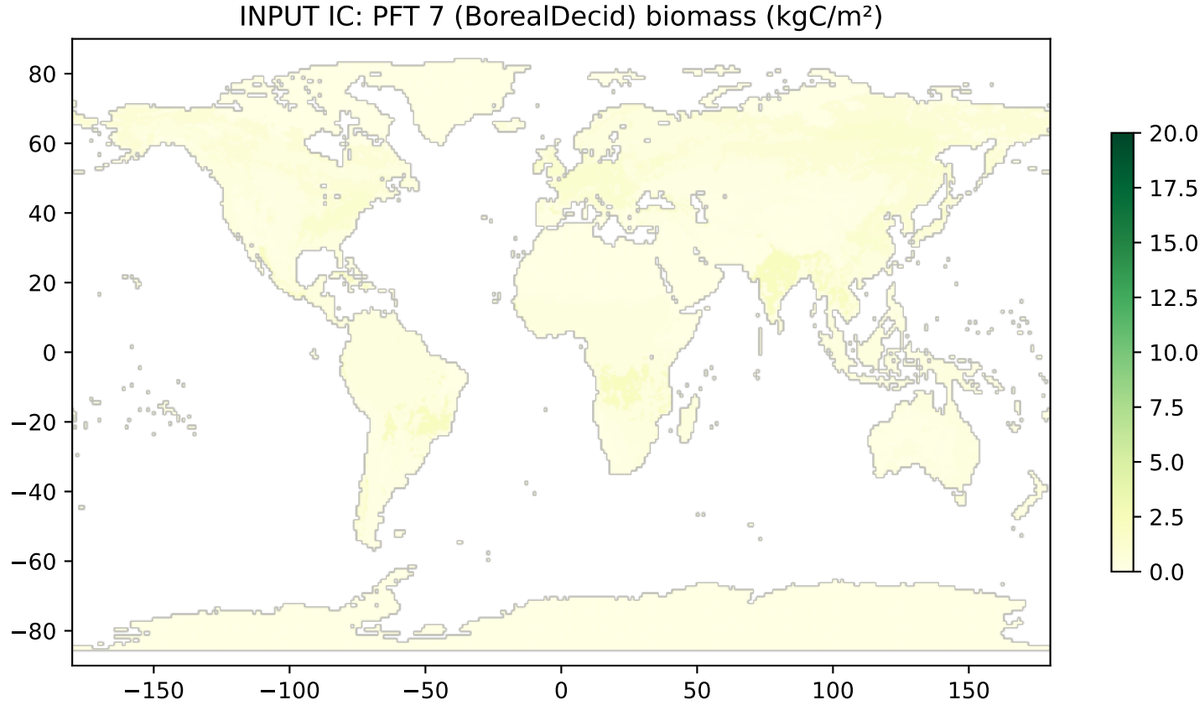


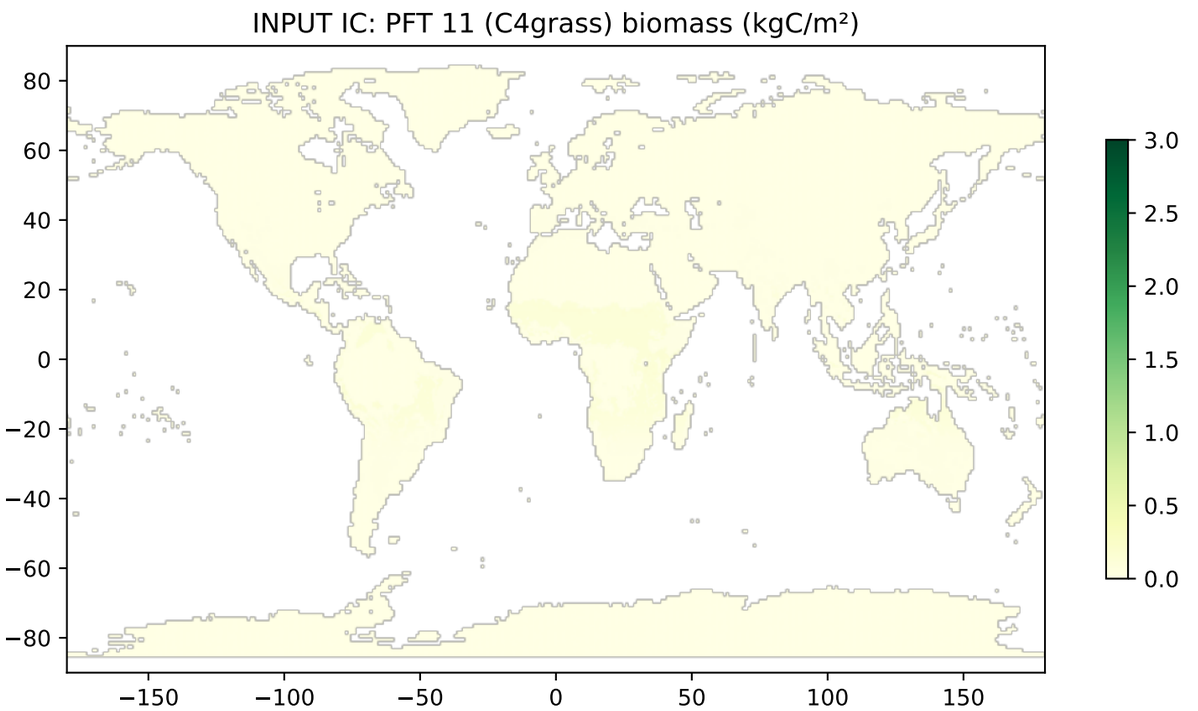
INPUTS: IC per-PFT biomass distribution (cold-start, exist=1 globally)

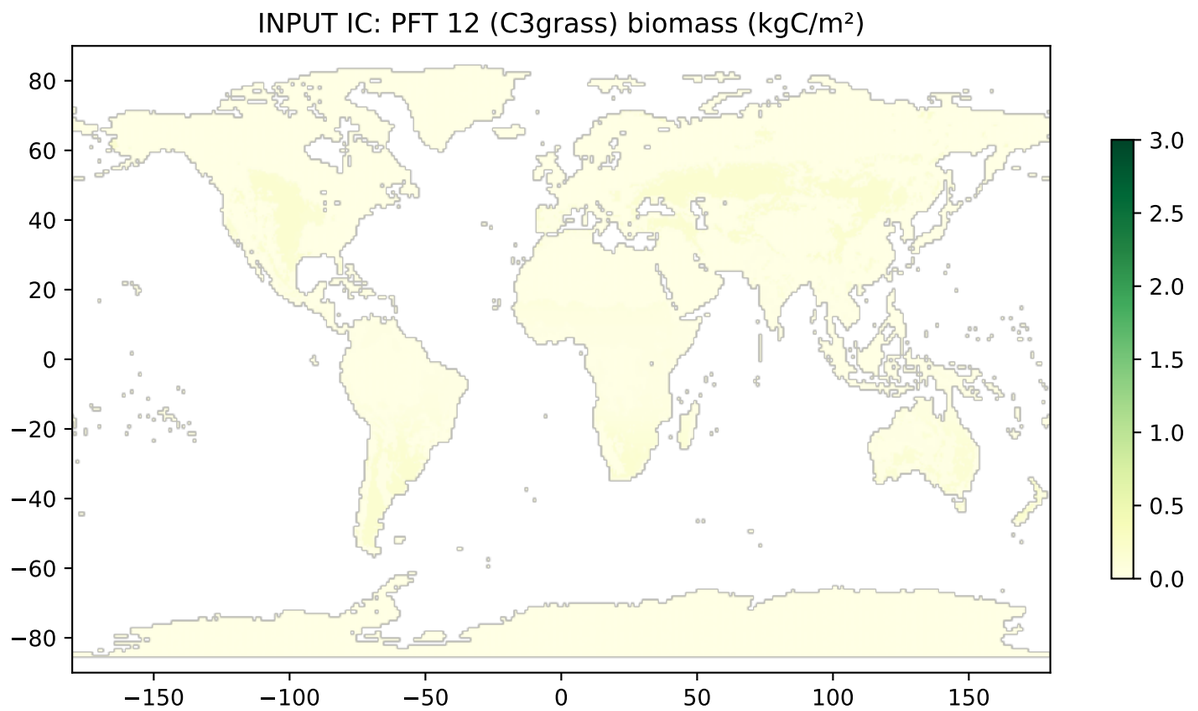




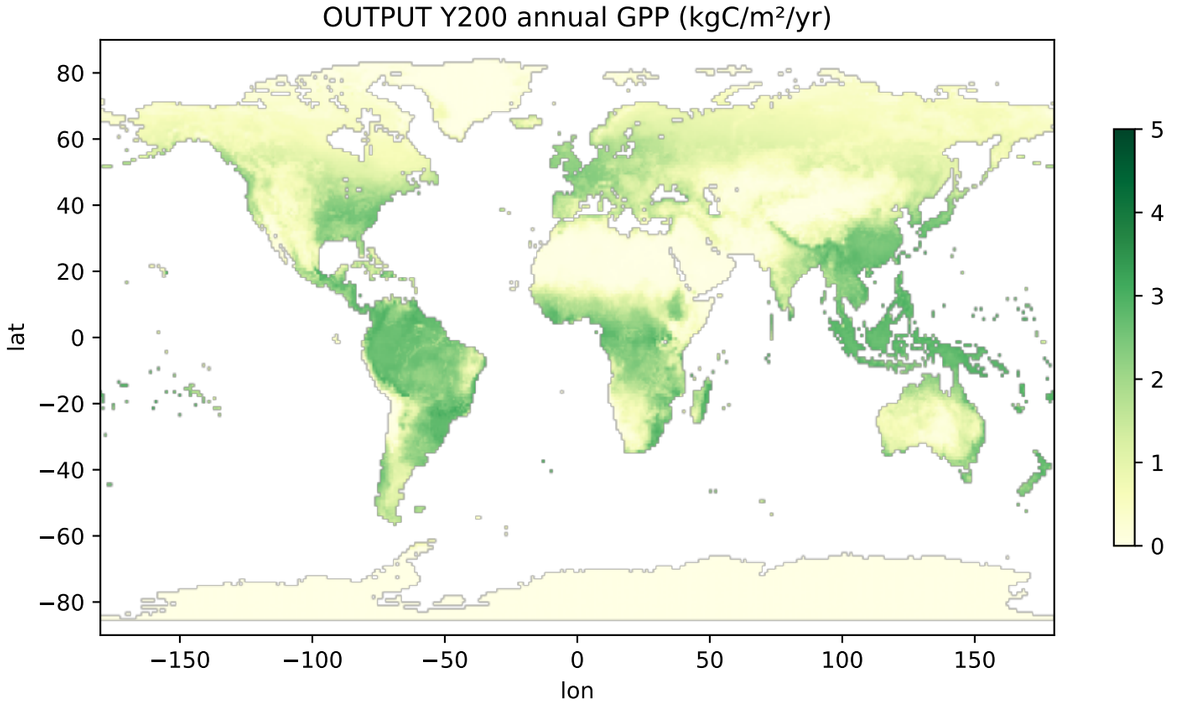


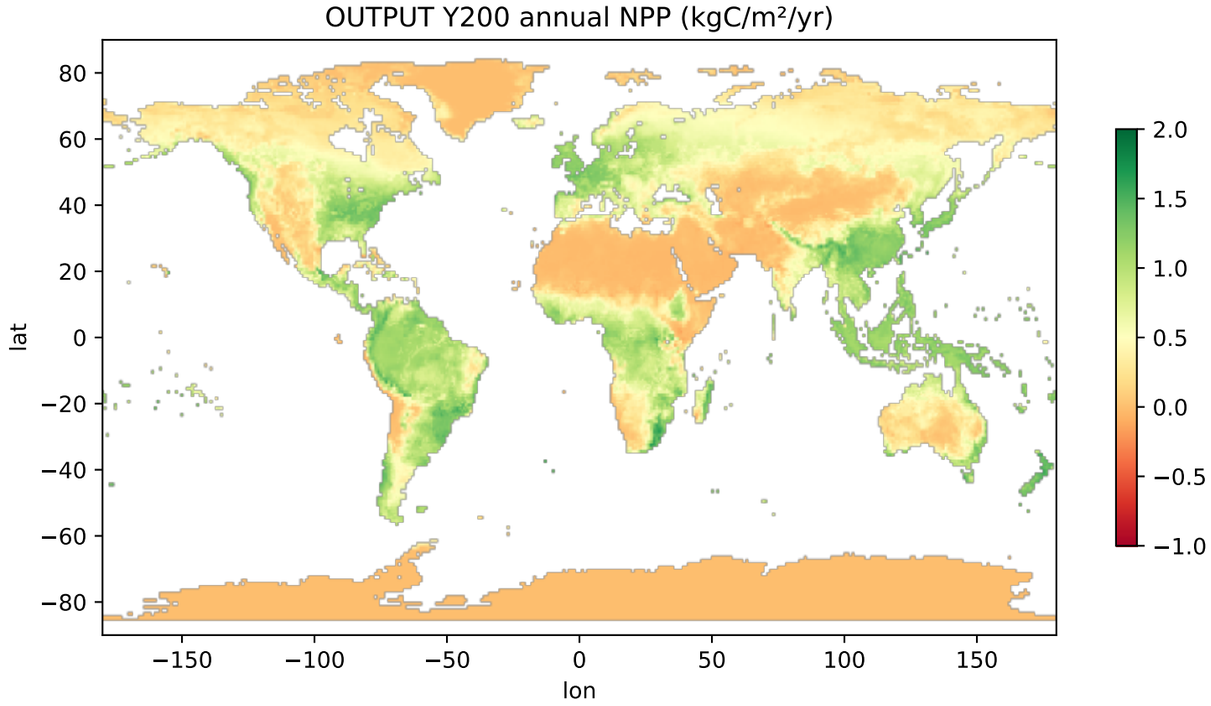


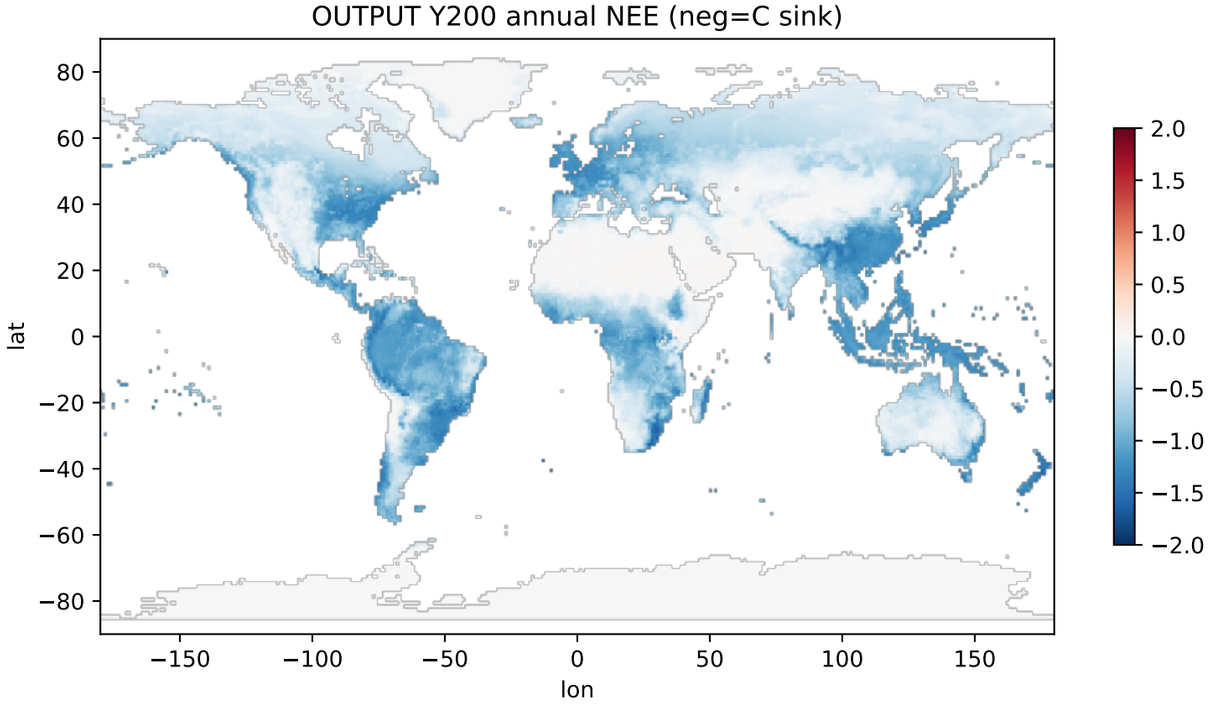


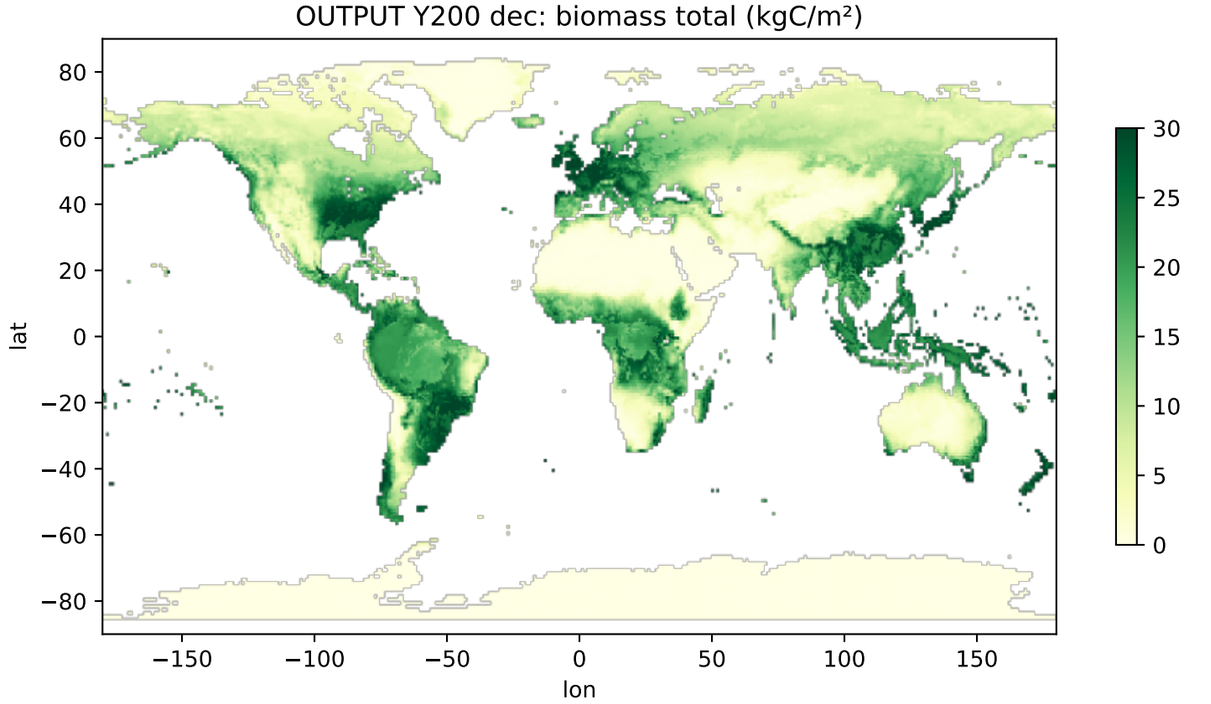


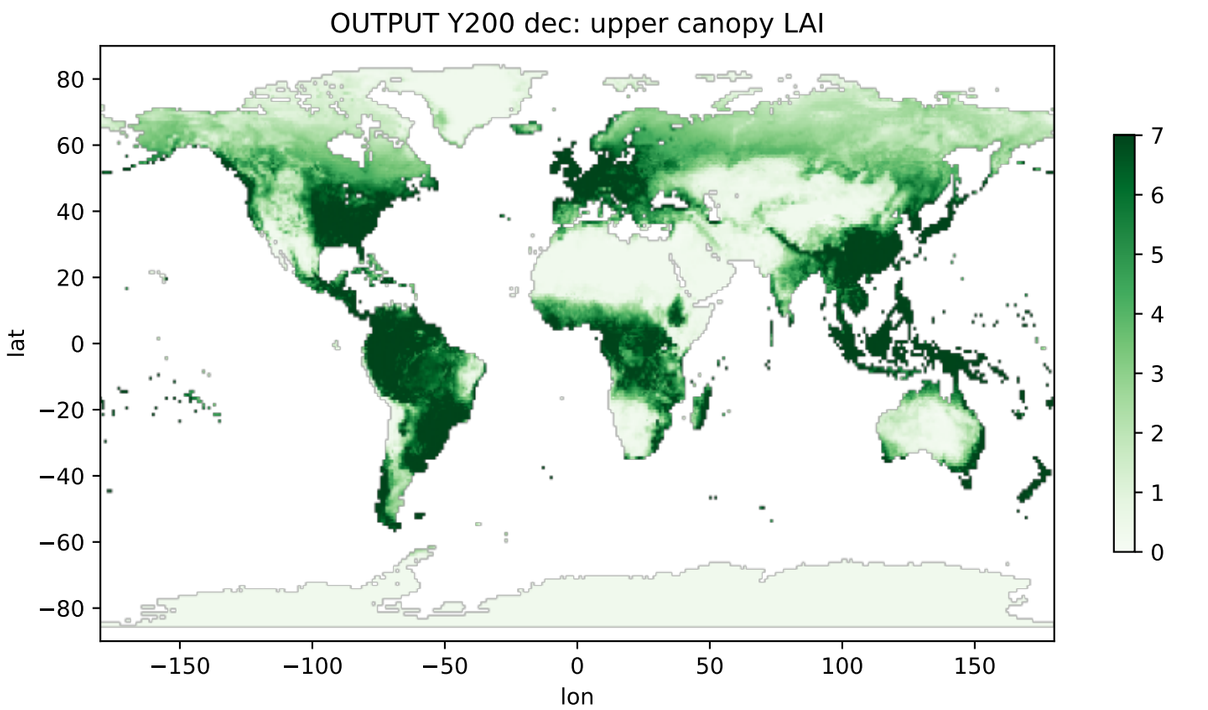
OUTPUTS YEAR 200: Carbon cycle

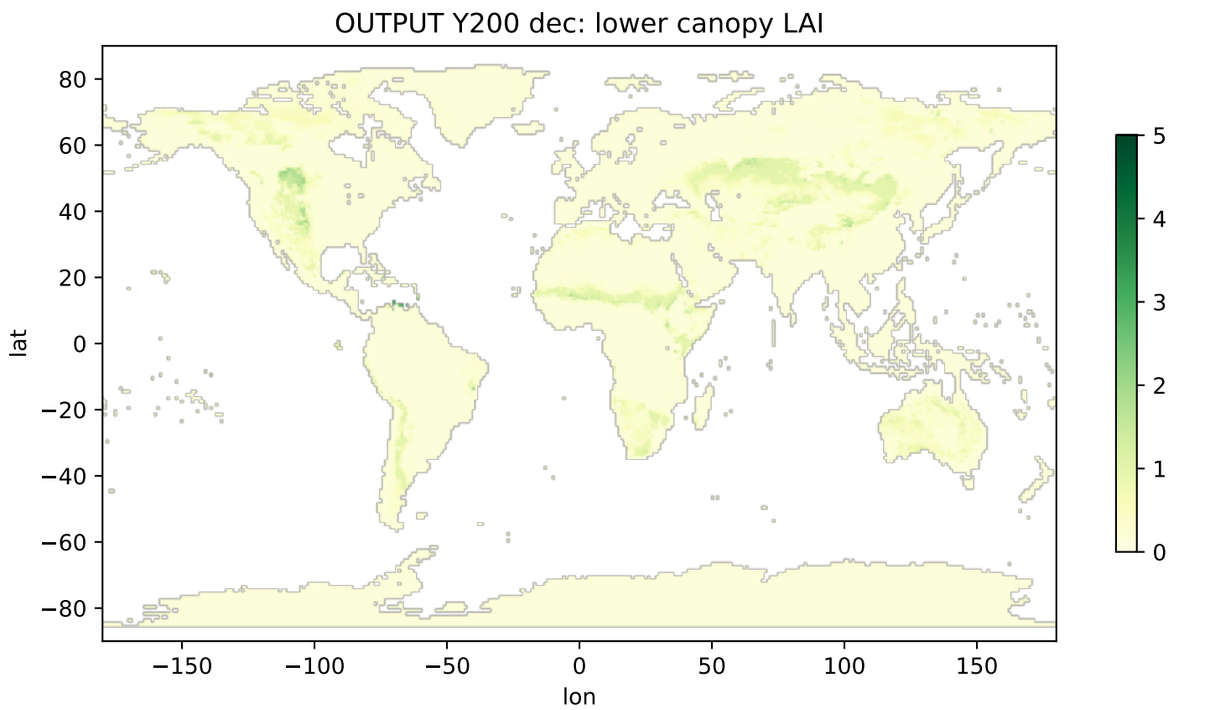






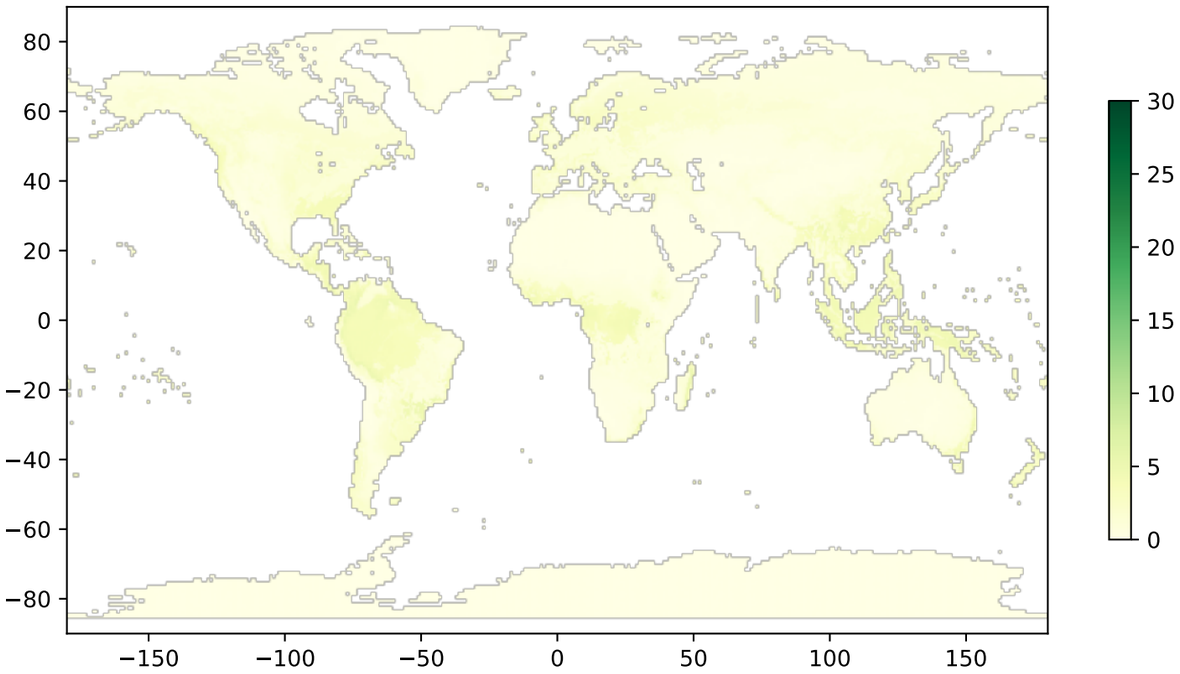




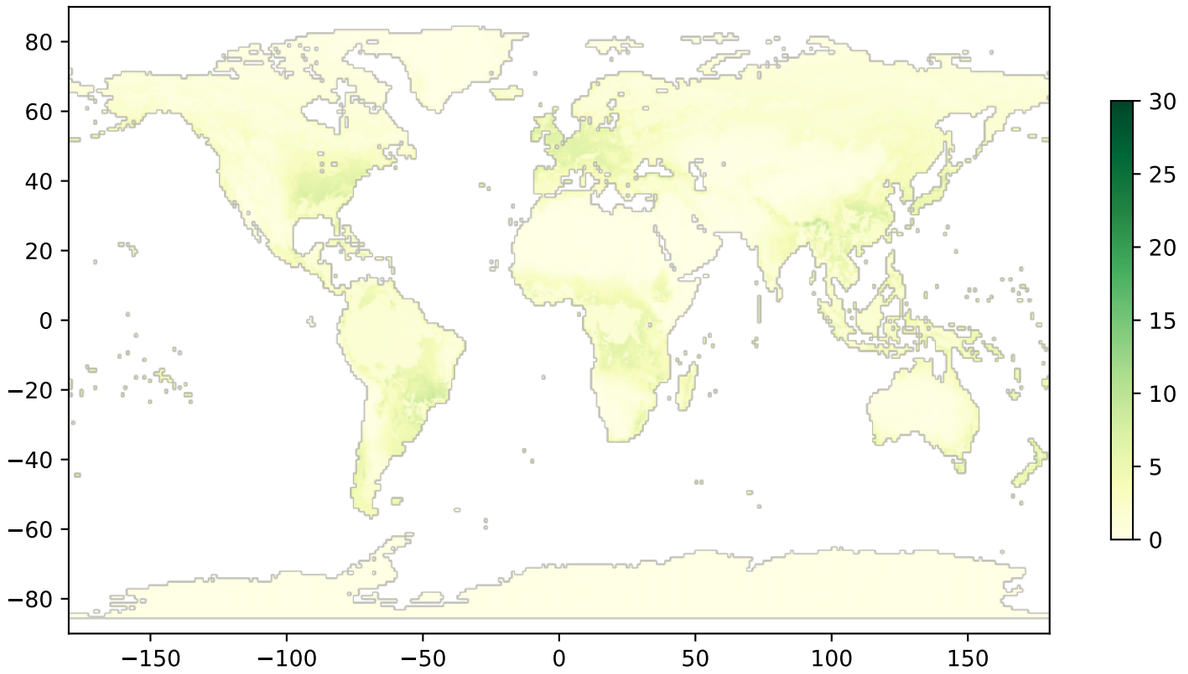


OUTPUTS Y200: per-PFT biomass (compare to IC)

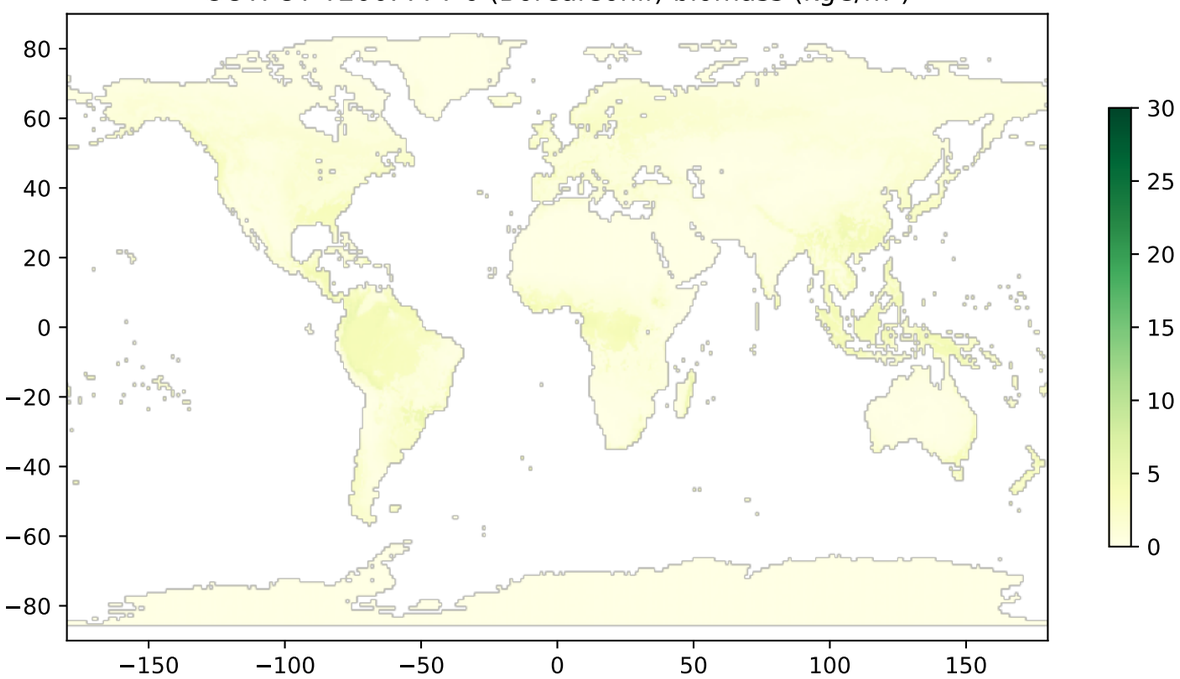
OUTPUT Y200: PFT 1 (TropEverGr) biomass (kgC/m²)



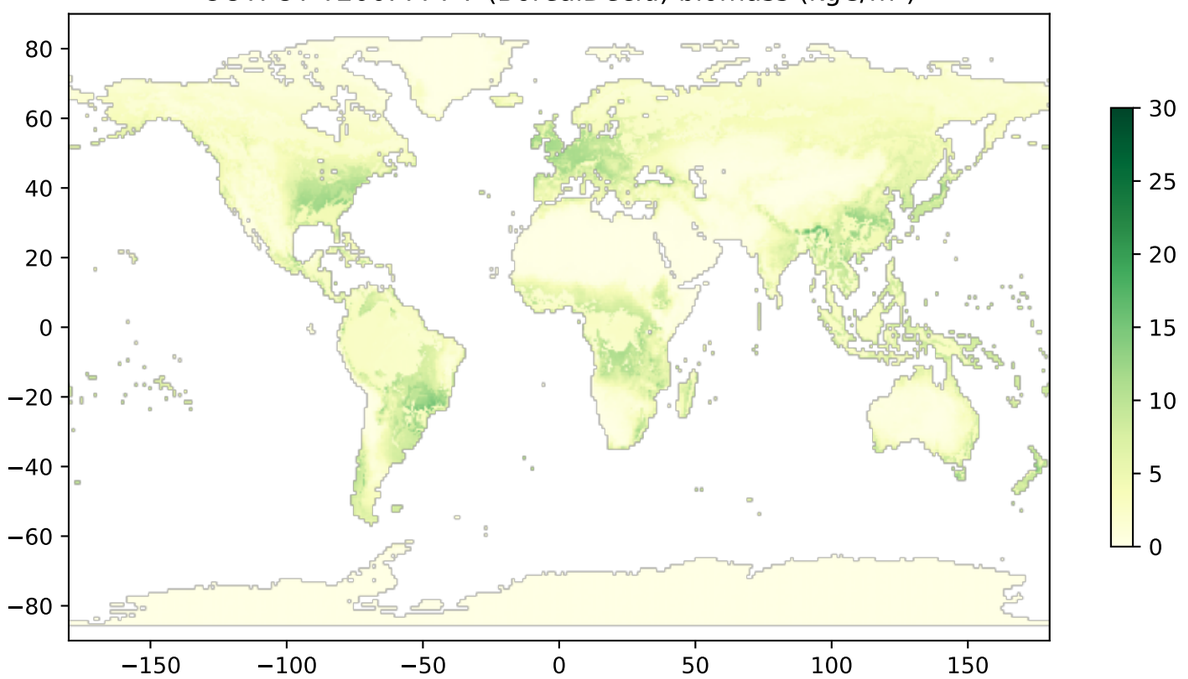
OUTPUT Y200: PFT 5 (TempDecid) biomass (kgC/m²)



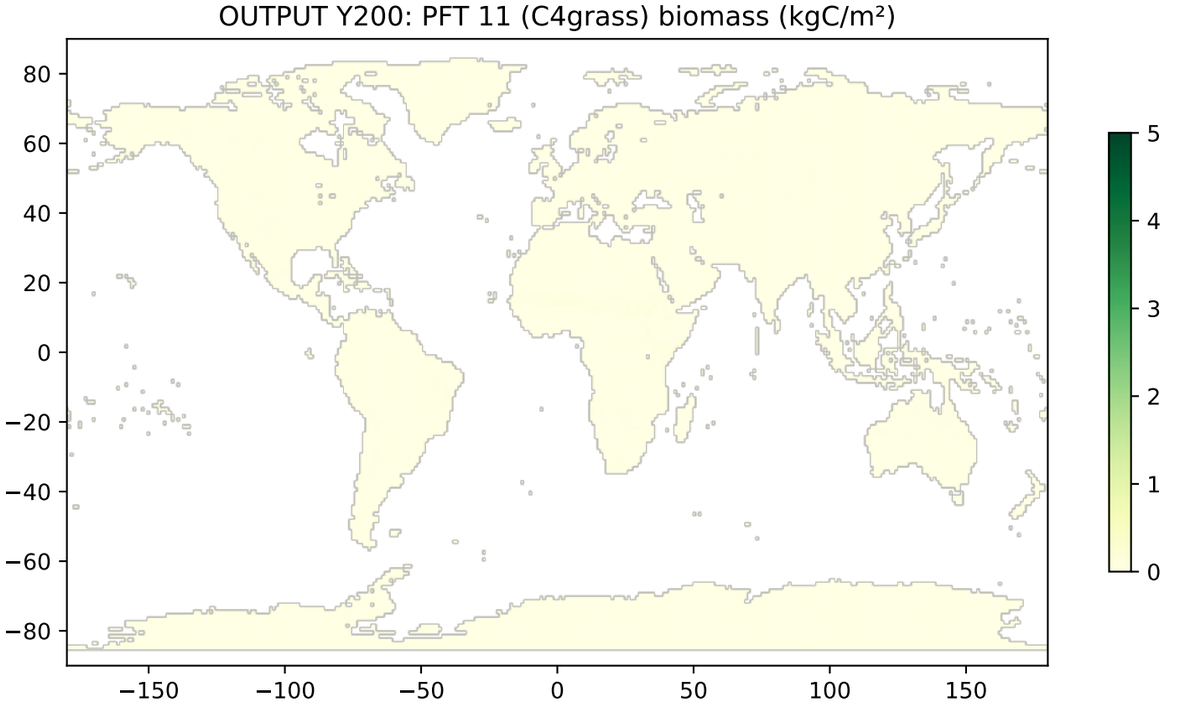
OUTPUT Y200: PFT 6 (BorealConif) biomass (kgC/m²)



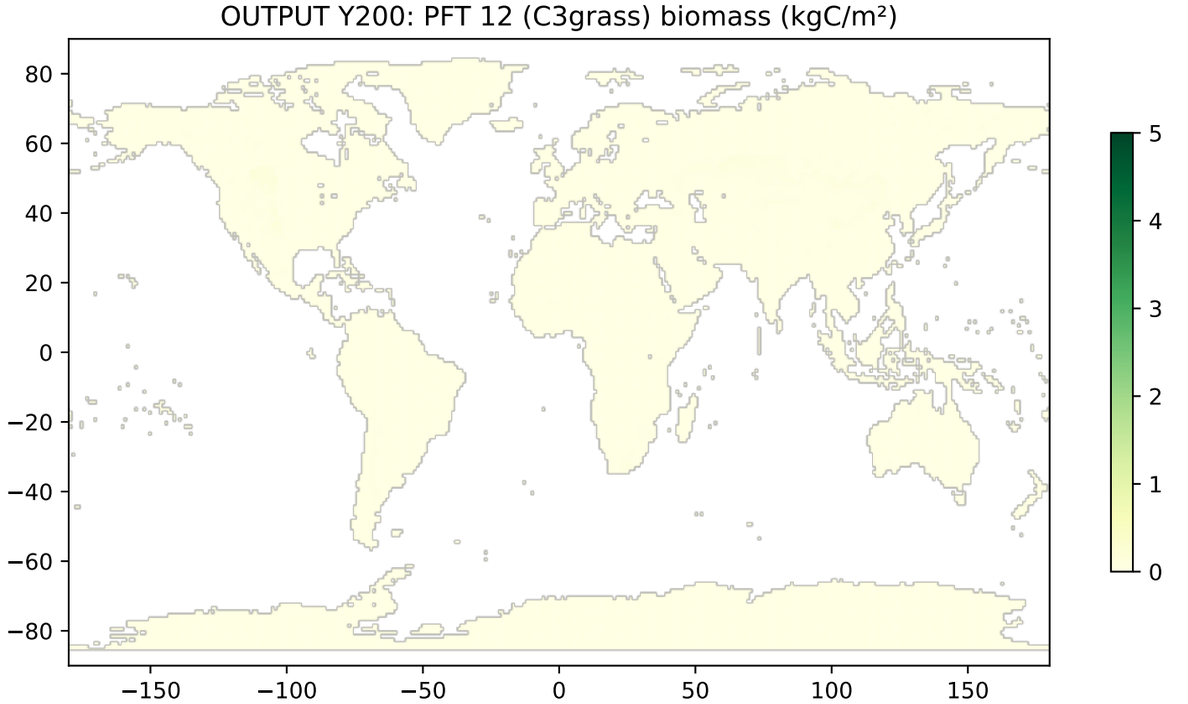
OUTPUT Y200: PFT 7 (BorealDecid) biomass (kgC/m²)

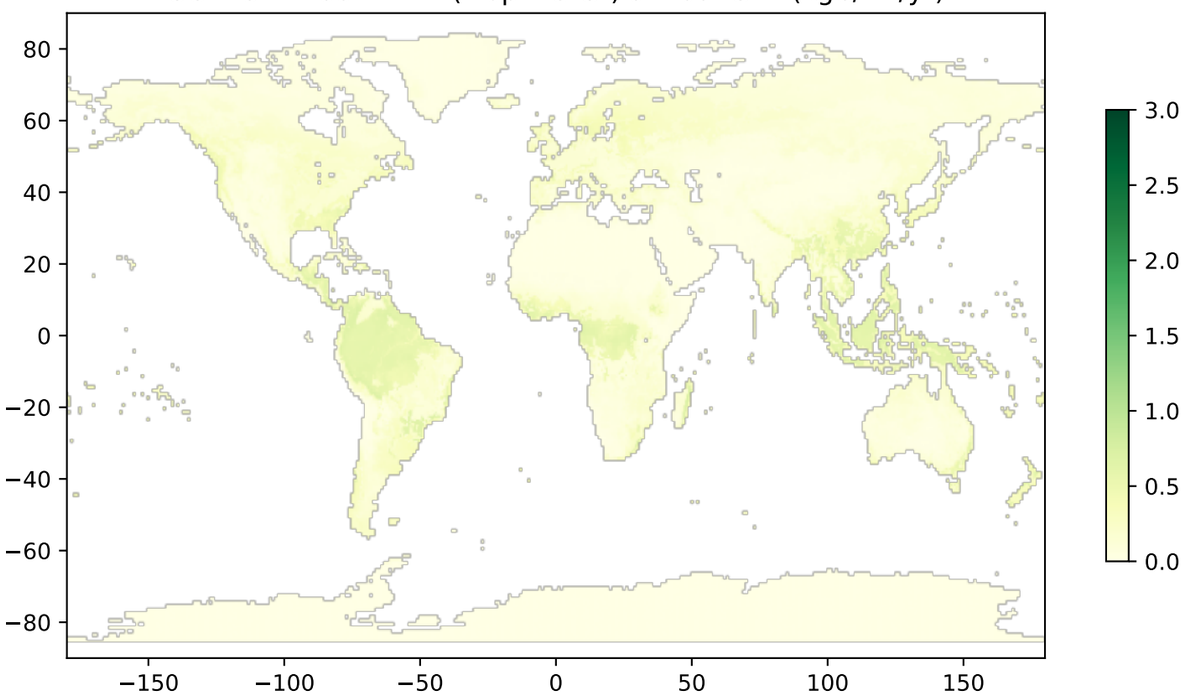


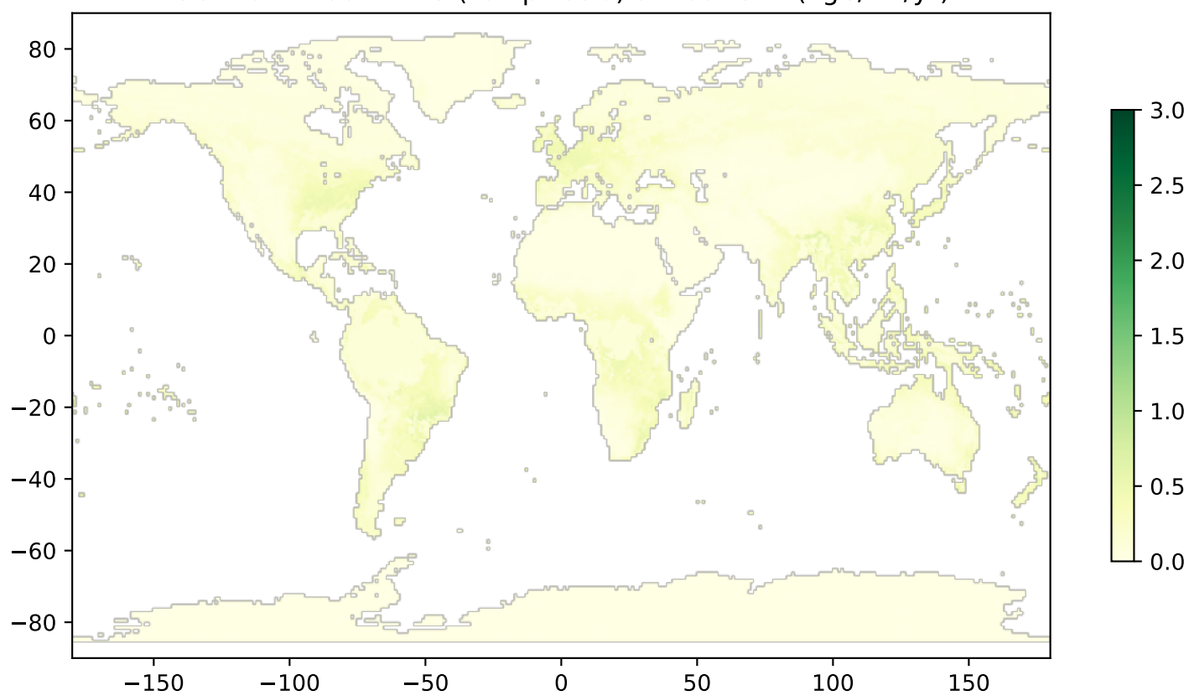
OUTPUT Y200: PFT 11 (C4grass) biomass (kgC/m²)

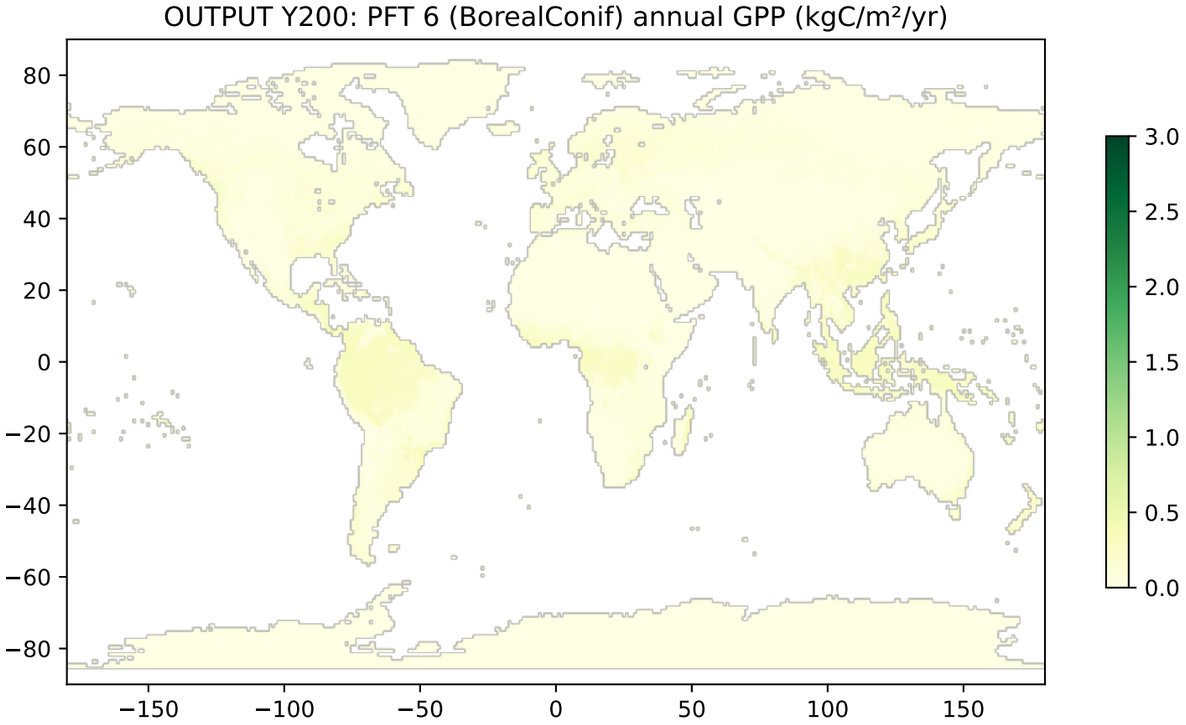


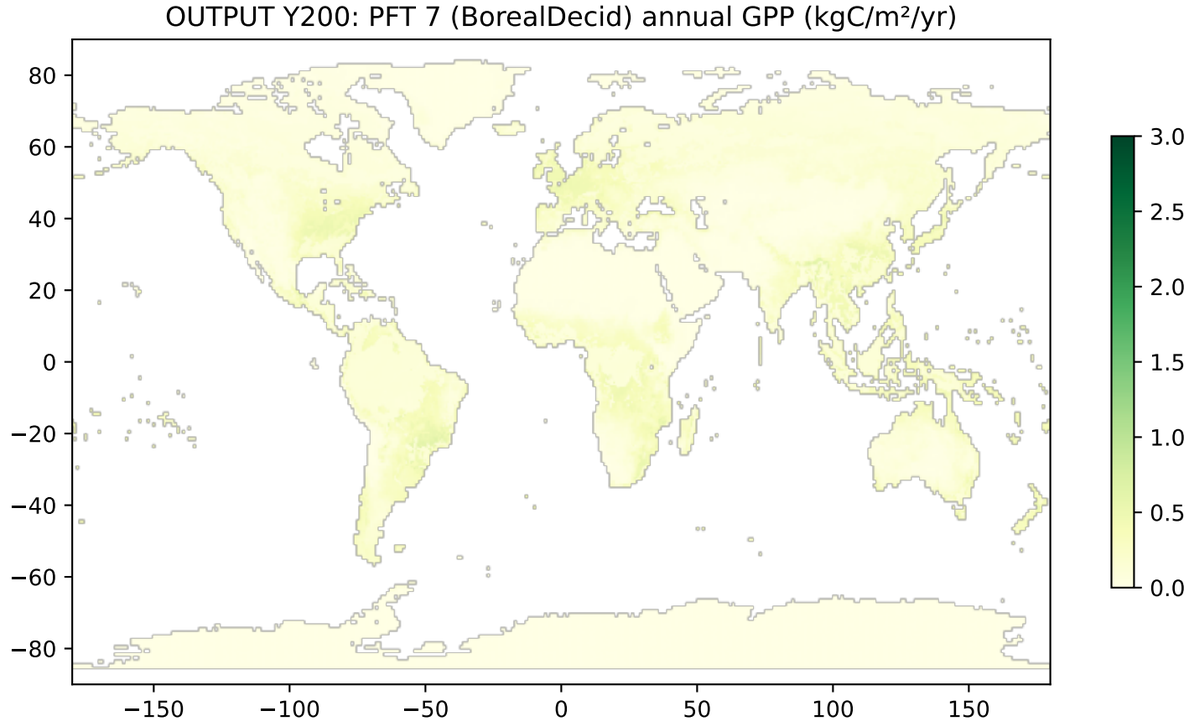
OUTPUT Y200: PFT 12 (C3grass) biomass (kgC/m²)

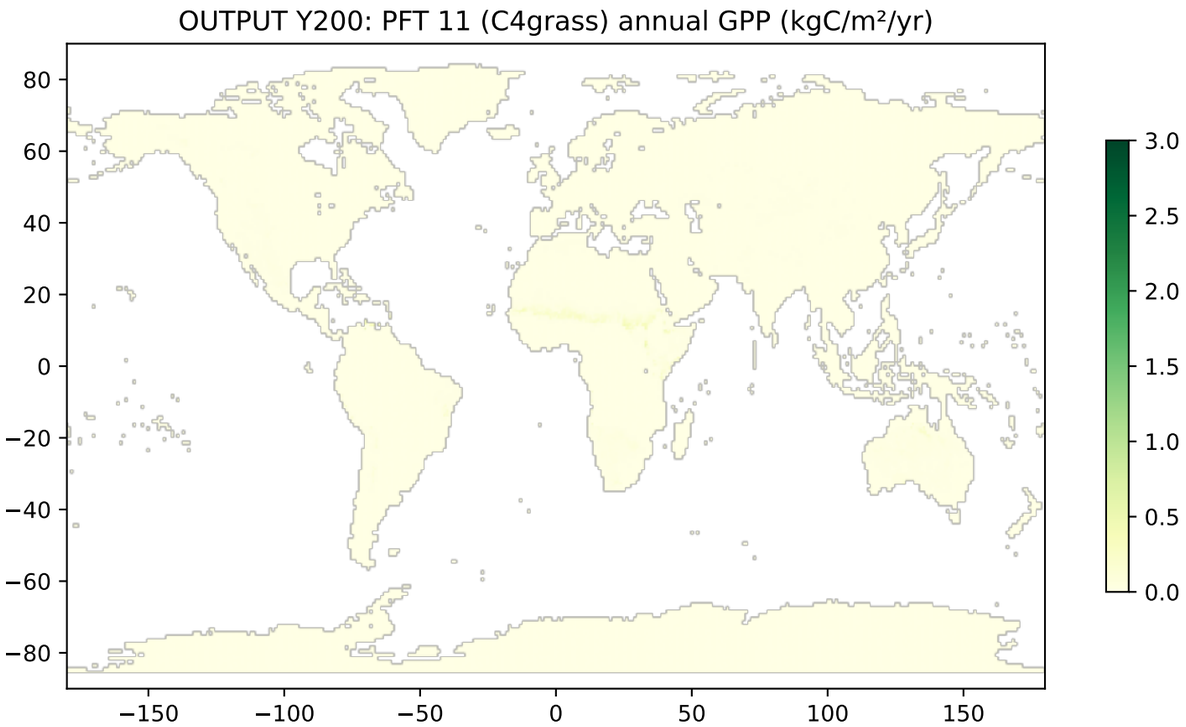


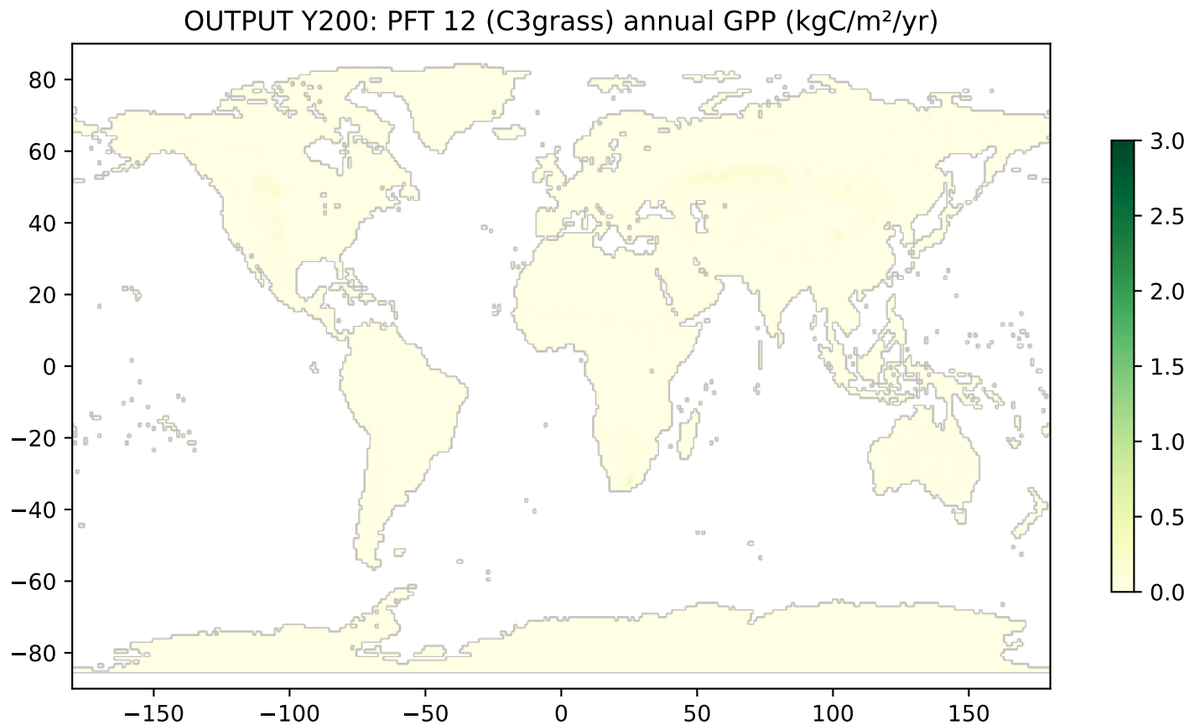

OUTPUT Y200: PFT 1 (TropEverGr) annual GPP (kgC/m²/yr)
80
60
40
20
0
−20
−40
−60
−80
−150
−100
−50
0
50
100
150
3.0
2.5
2.0
1.5
1.0
0.5
0.0


OUTPUT Y200: PFT 5 (TempDecid) annual GPP (kgC/m²/yr)
80
60
40
20
0
−20
−40
−60
−80
−150
−100
−50
0
50
100
150
3.0
2.5
2.0
1.5
1.0
0.5
0.0

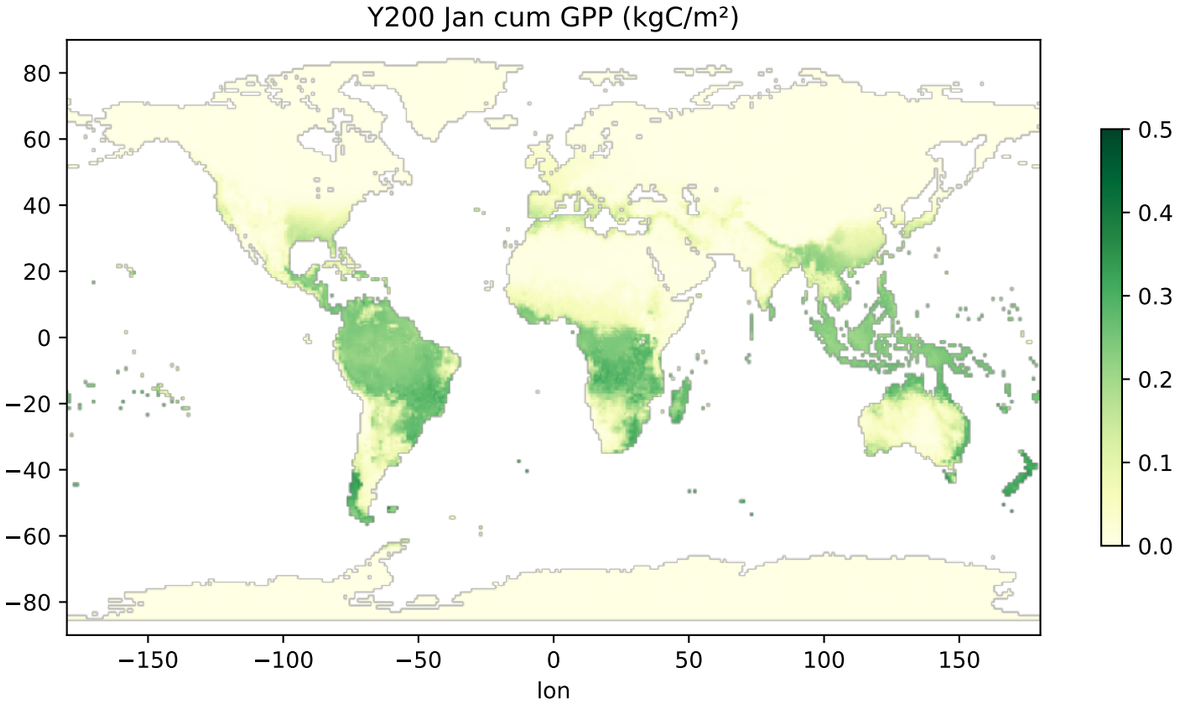

OUTPUT Y200: PFT 6 (BorealConif) annual GPP (kgC/m²/yr)
80
60
40
20
0
−20
−40
−60
−80
−150
−100
−50
0
50
100
150
3.0
2.5
2.0
1.5
1.0
0.5
0.0

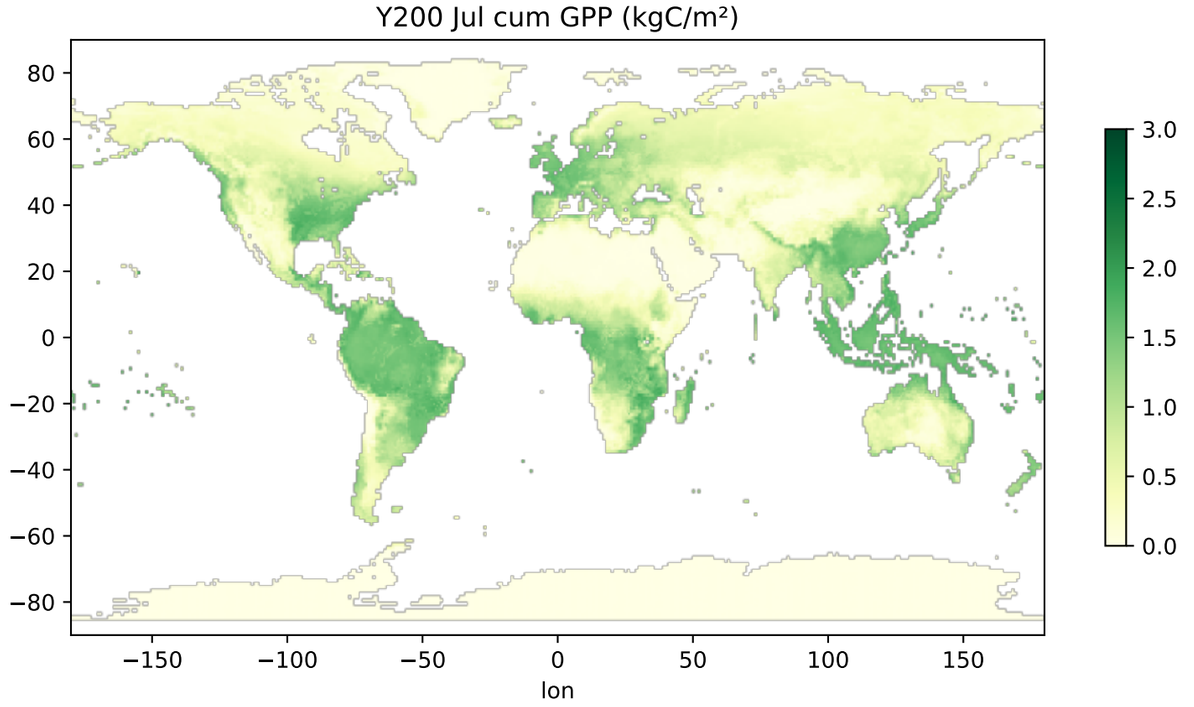

OUTPUT Y200: PFT 7 (BorealDecid) annual GPP (kgC/m²/yr)
80
60
40
20
0
−20
−40
−60
−80
−150
−100
−50
0
50
100
150
3.0
2.5
2.0
1.5
1.0
0.5
0.0

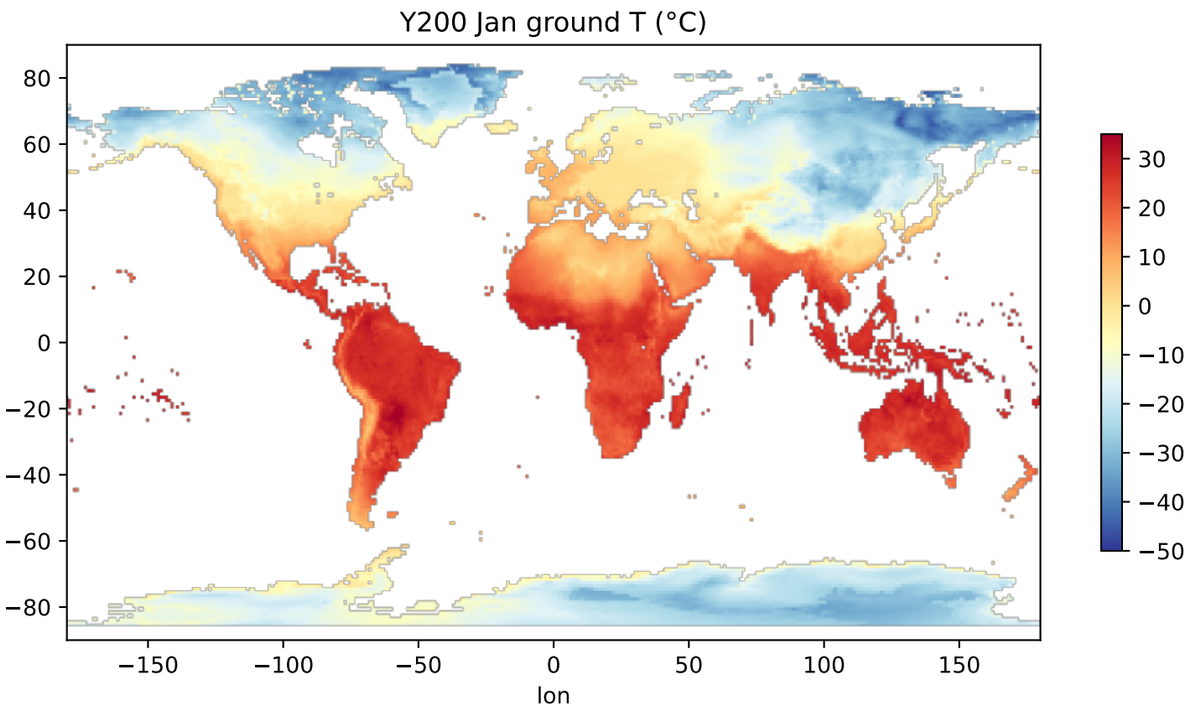

OUTPUT Y200: PFT 11 (C4grass) annual GPP (kgC/m²/yr)
80
60
40
20
0
−20
−40
−60
−80
−150
−100
−50
0
50
100
150
3.0
2.5
2.0
1.5
1.0
0.5
0.0

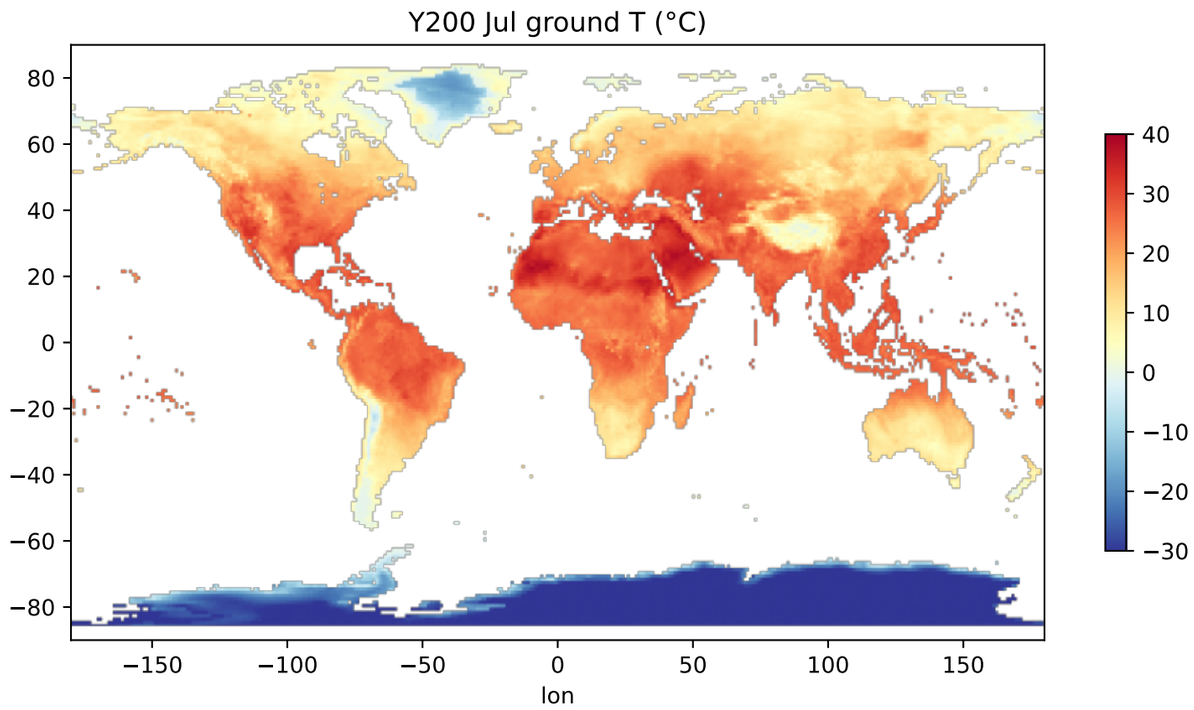

OUTPUT Y200: PFT 12 (C3grass) annual GPP (kgC/m²/yr)
80
60
40
20
0
−20
−40
−60
−80
−150
−100
−50
0
50
100
150
3.0
2.5
2.0
1.5
1.0
0.5
0.0

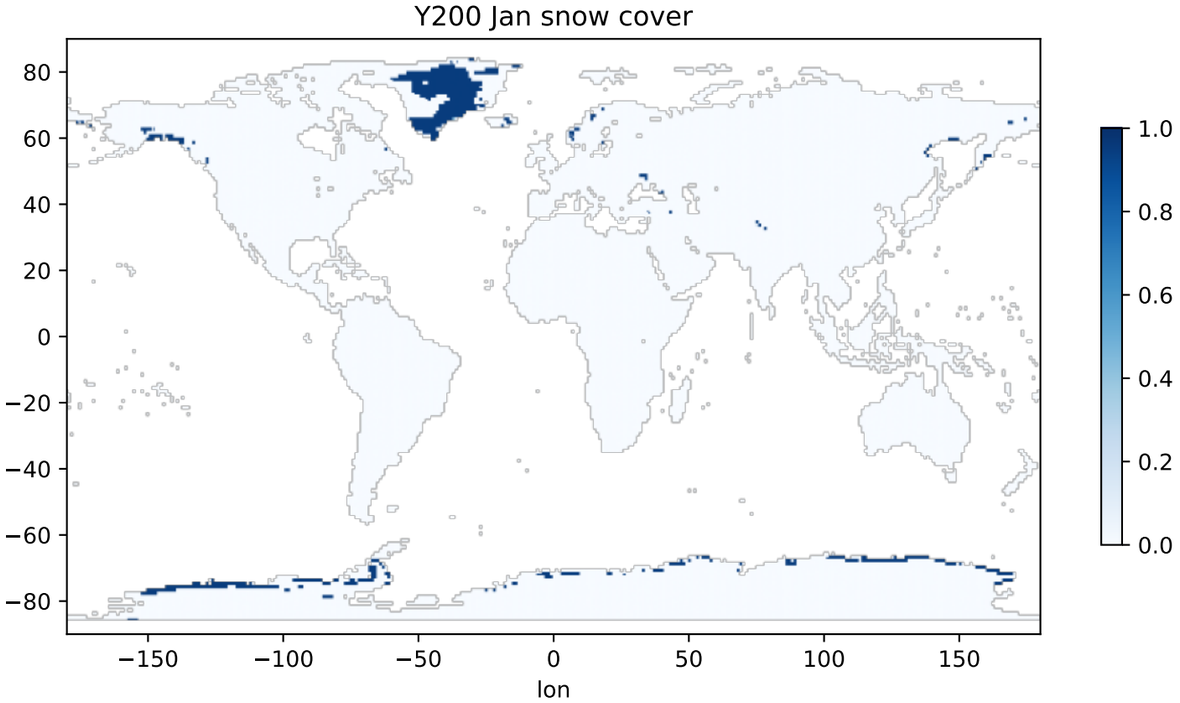
OUTPUTS Y200: Seasonal contrast

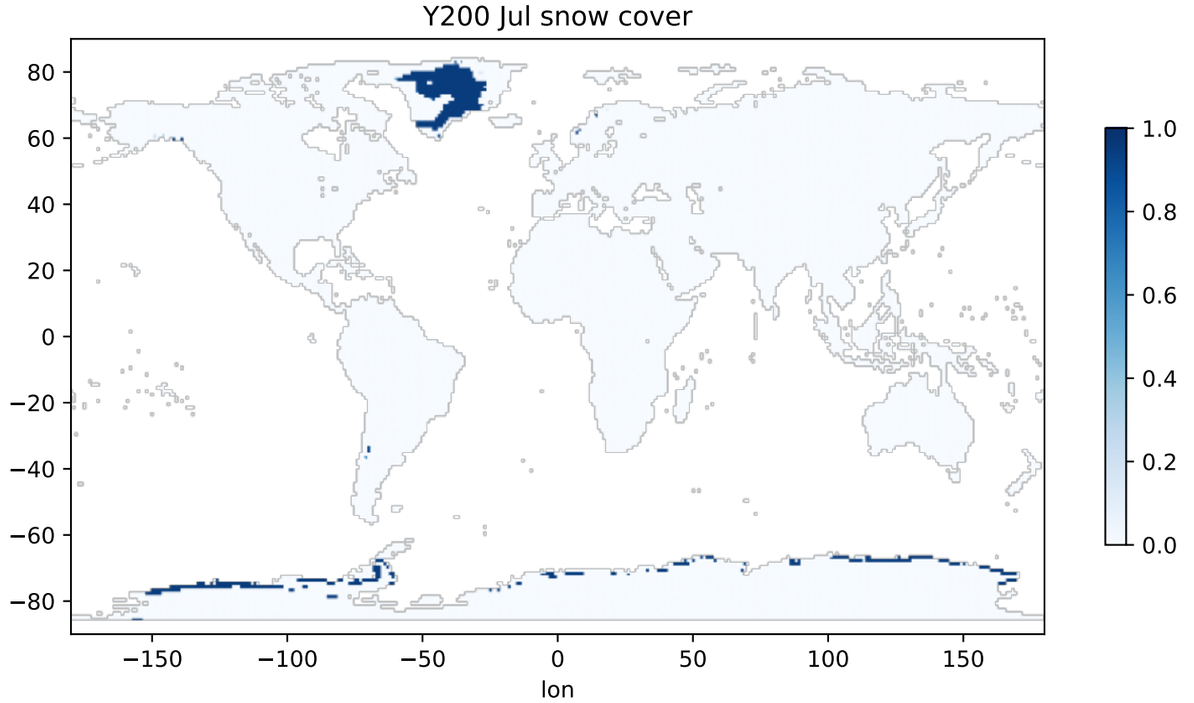




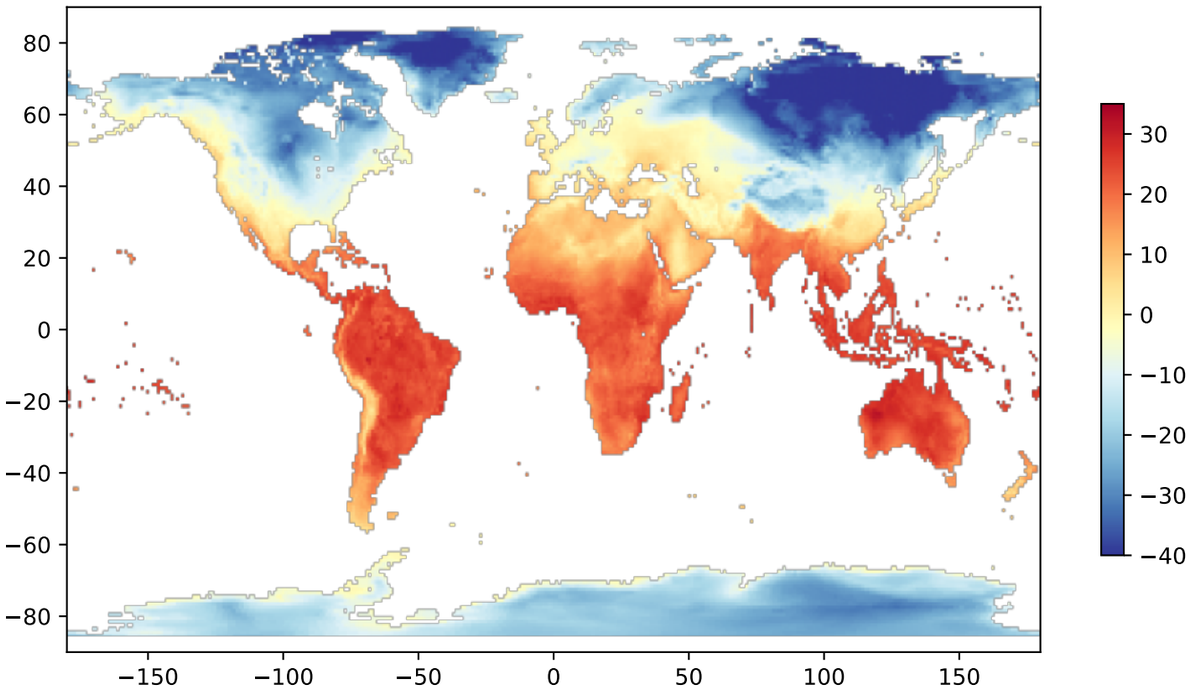




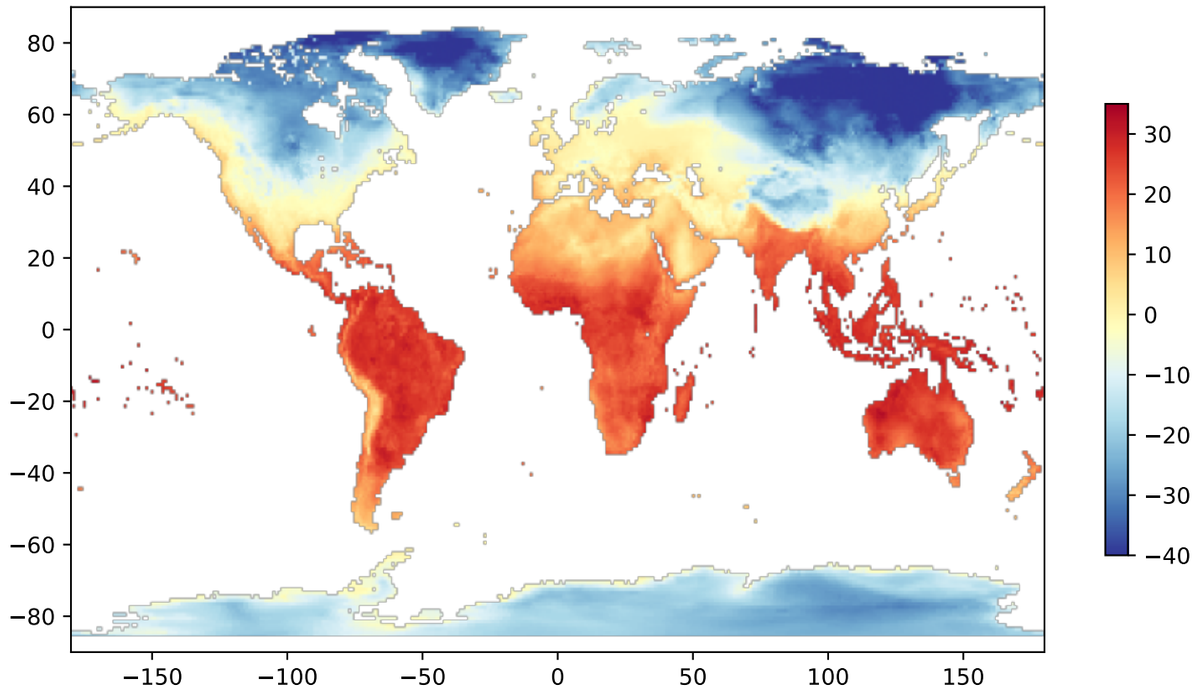




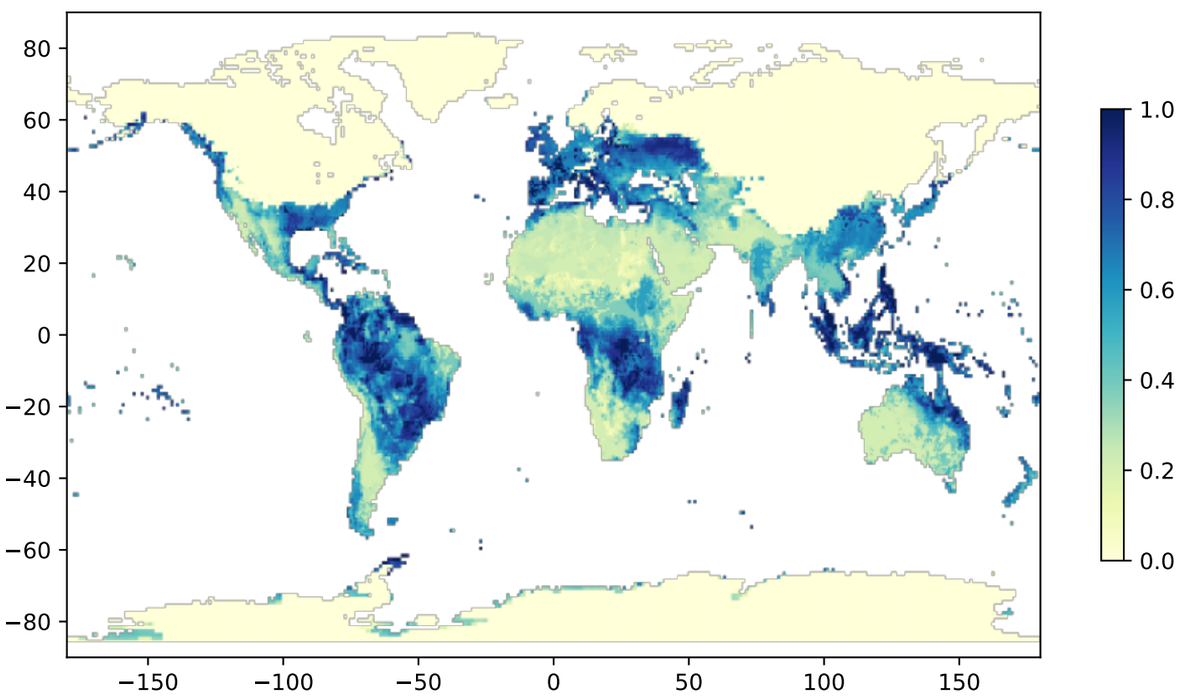
Y200 Dec upper canopy T (°C)



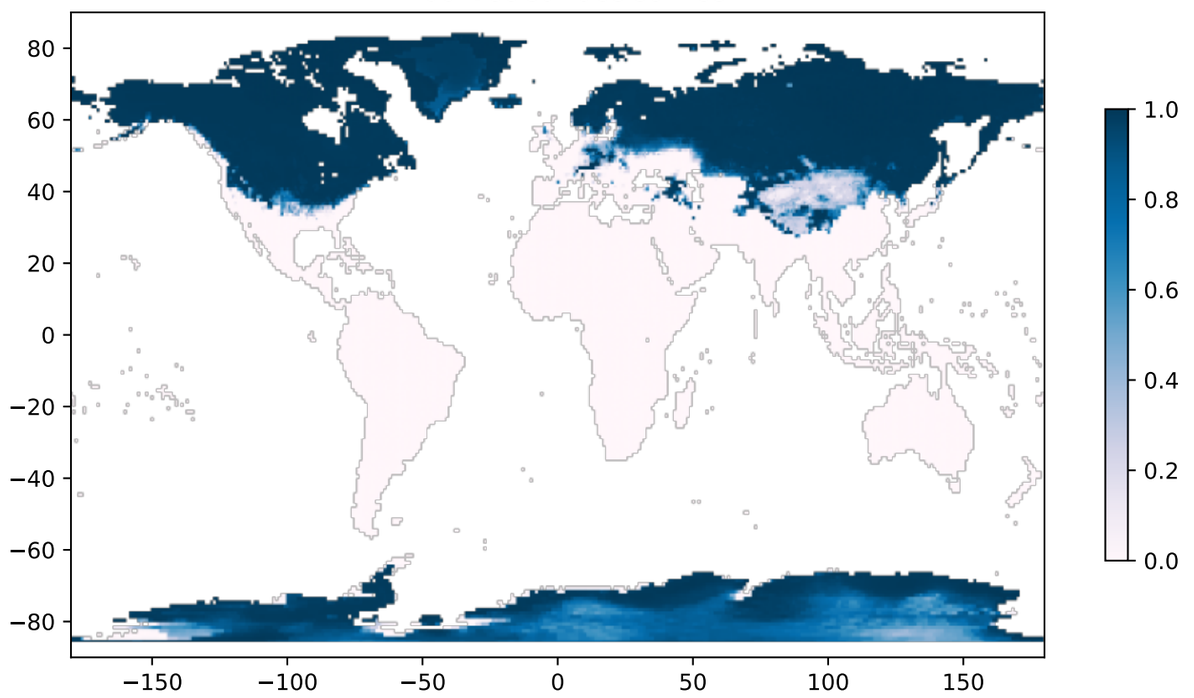
Y200 Dec lower canopy T (°C)



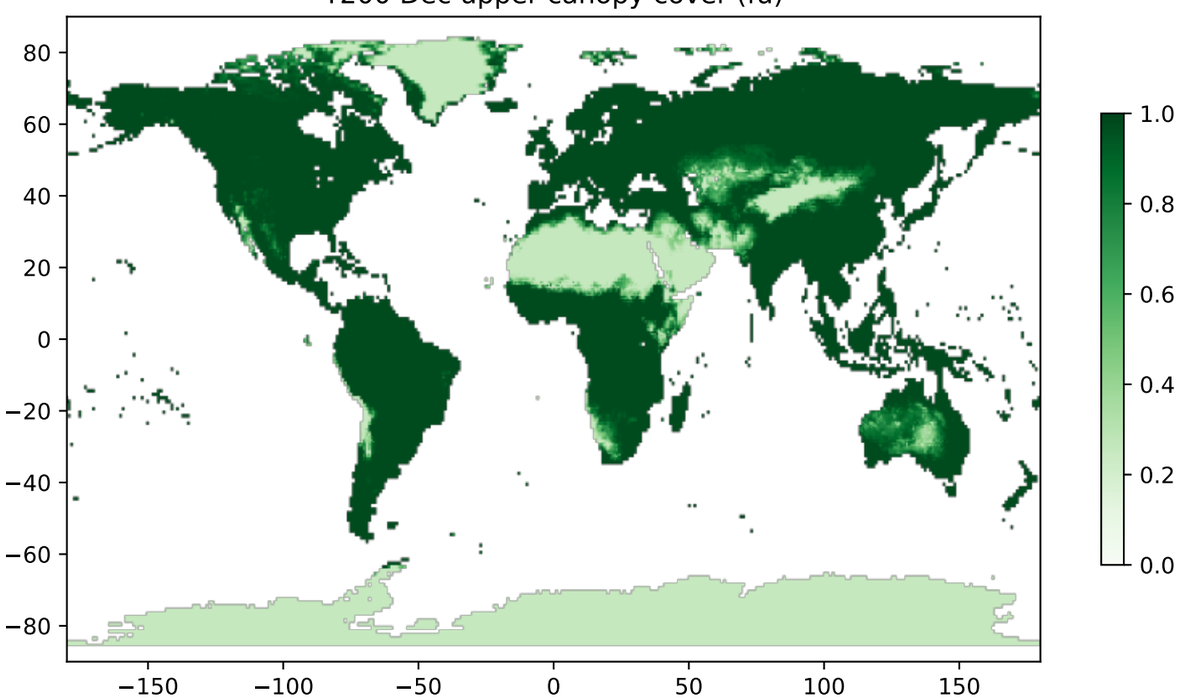
Y200 Dec surface soil moisture



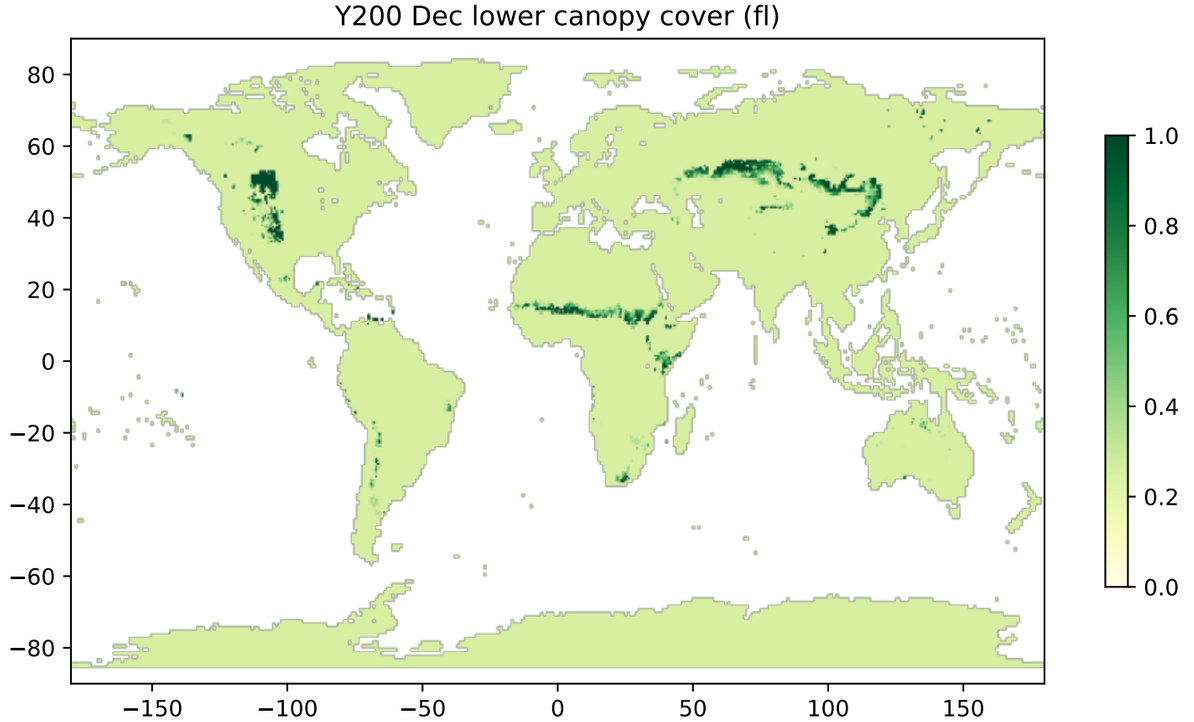
Y200 Dec surface soil ice



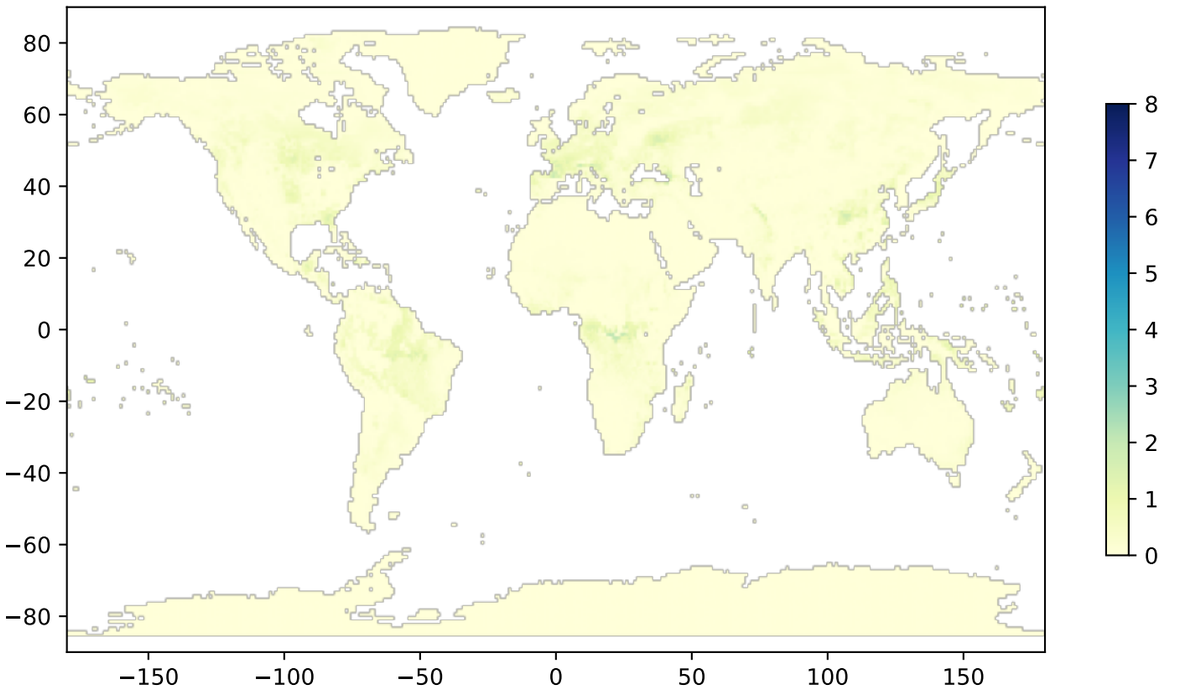
Y200 Dec upper canopy cover (fu)



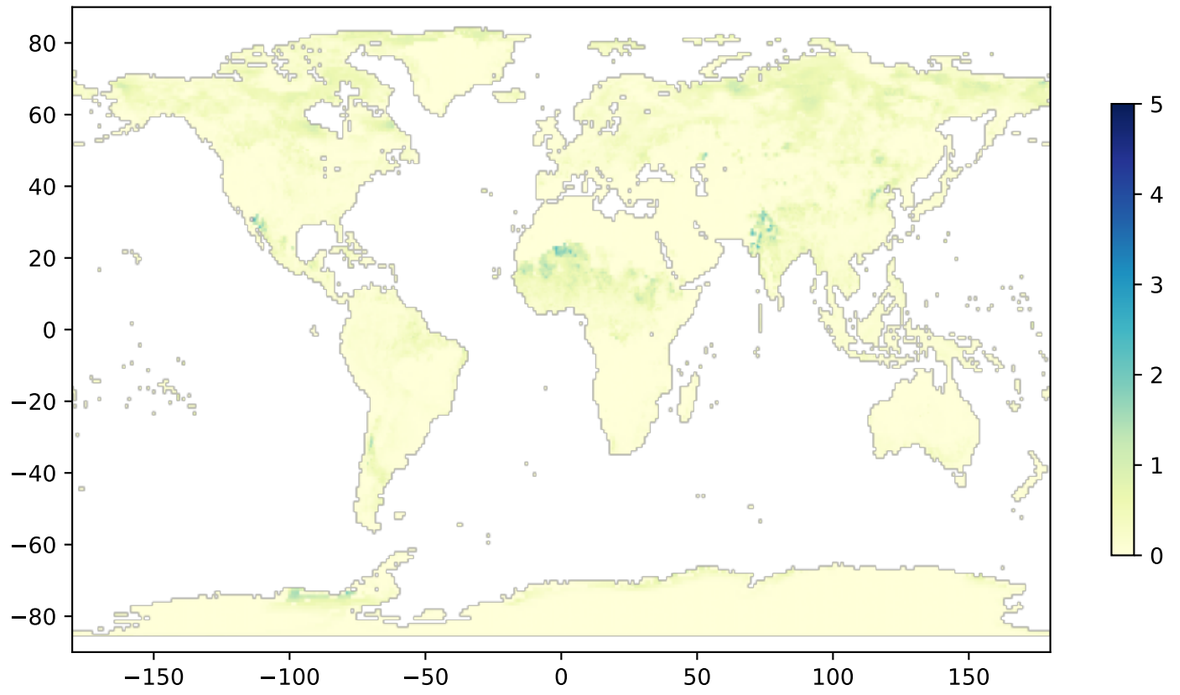
Y200 Dec lower canopy cover (fl)



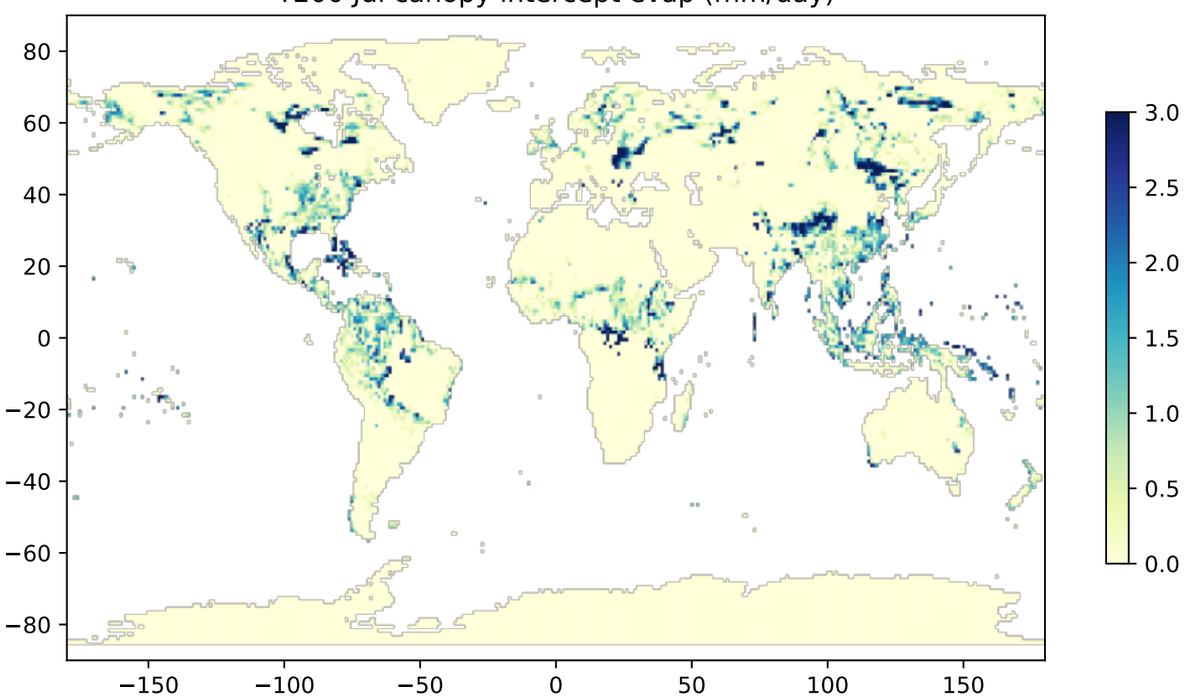
Y200 Jul transpiration (mm/day)



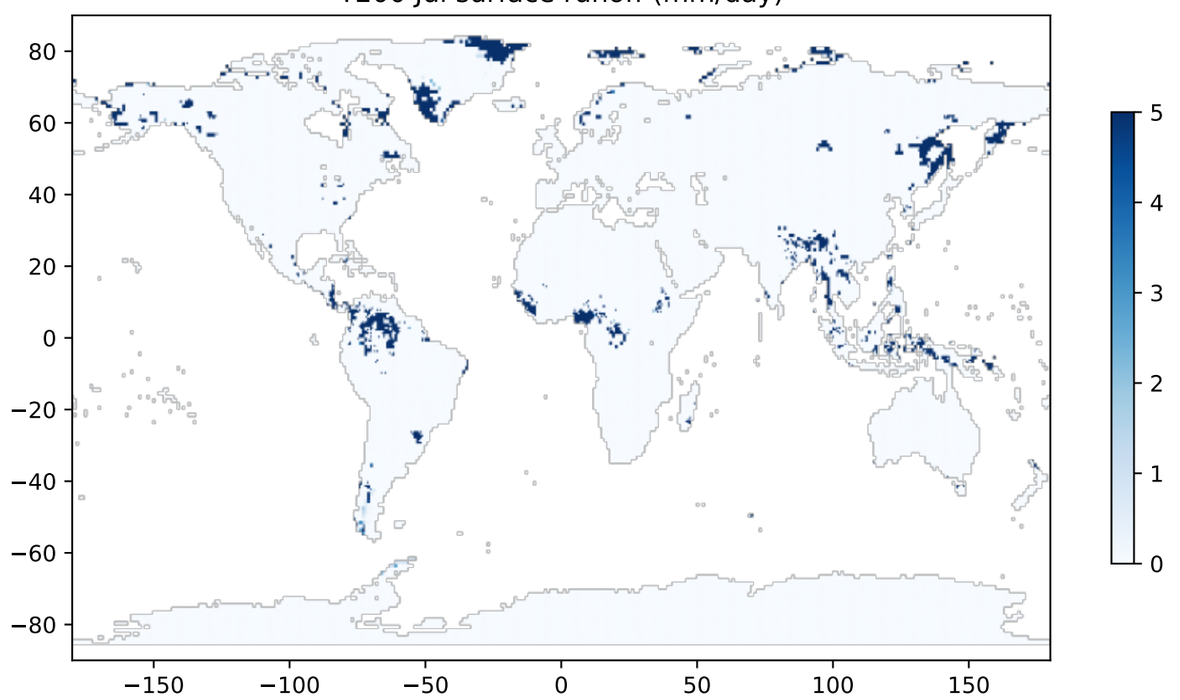
Y200 Jul soil evap (mm/day)



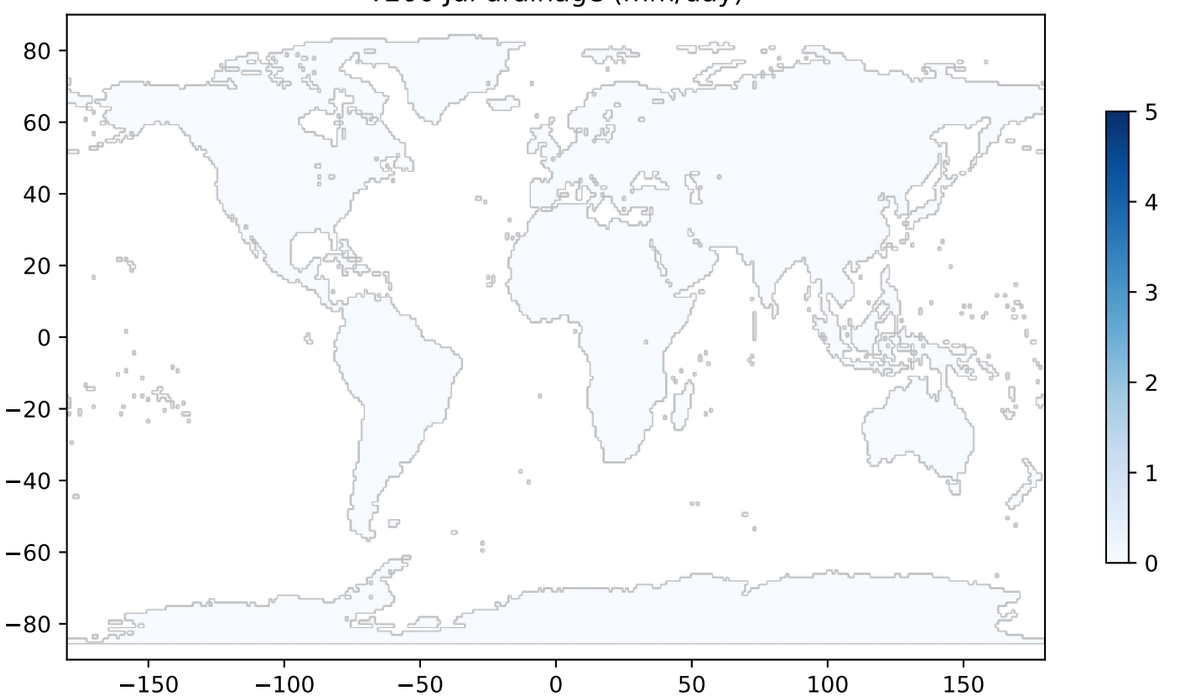
Y200 Jul canopy intercept evap (mm/day)



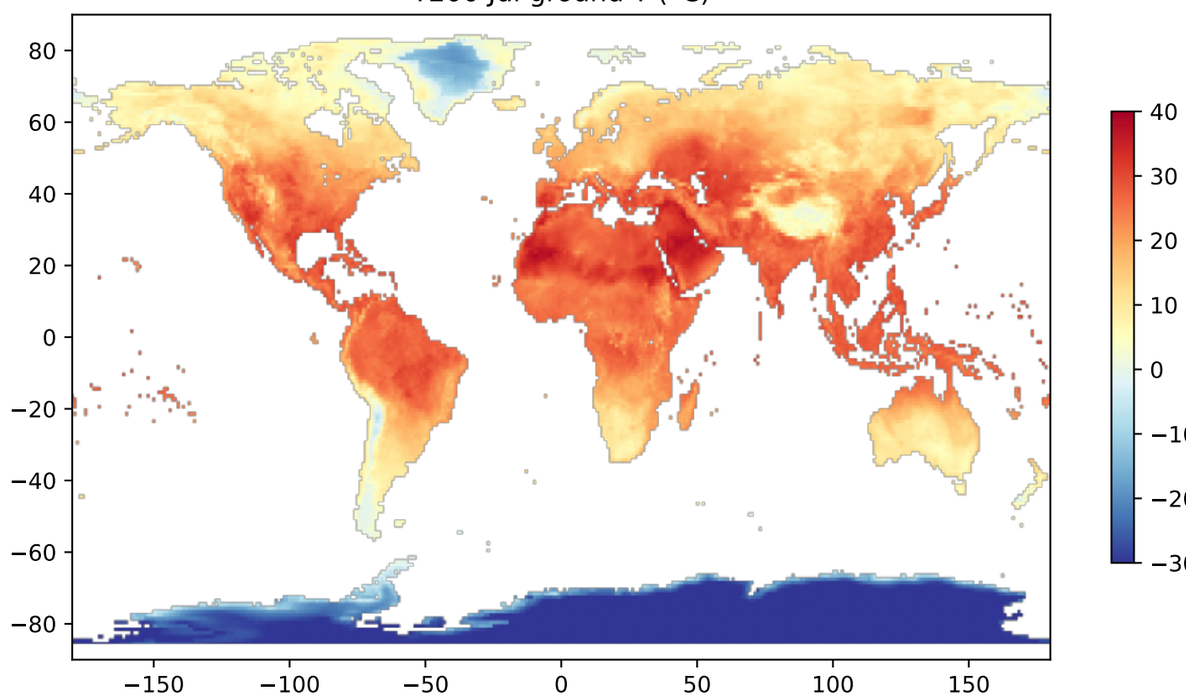
Y200 Jul surface runoff (mm/day)

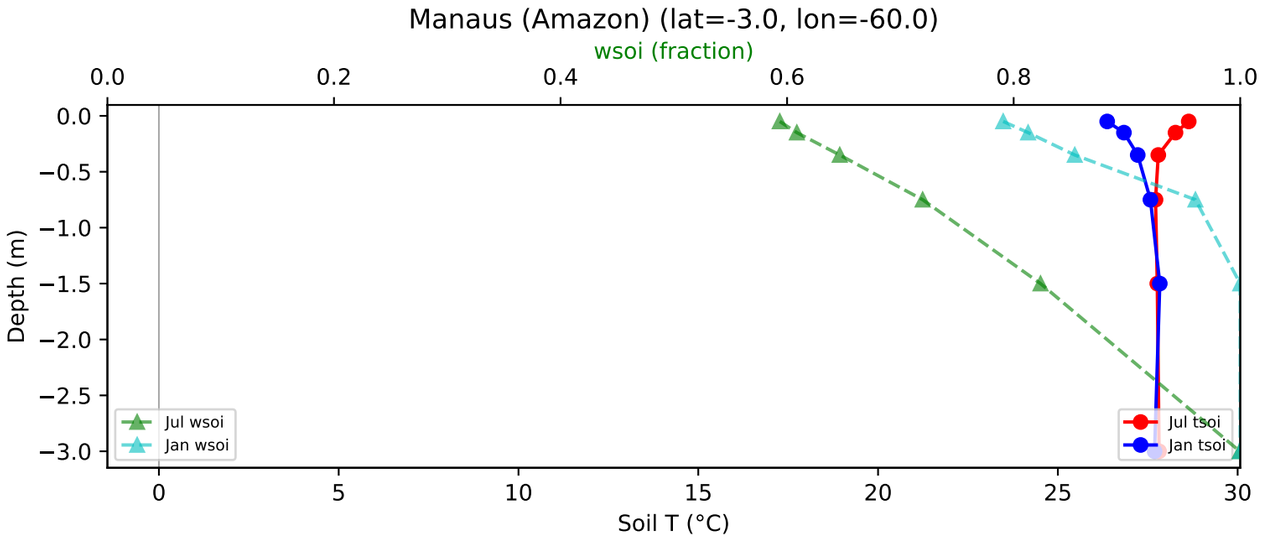


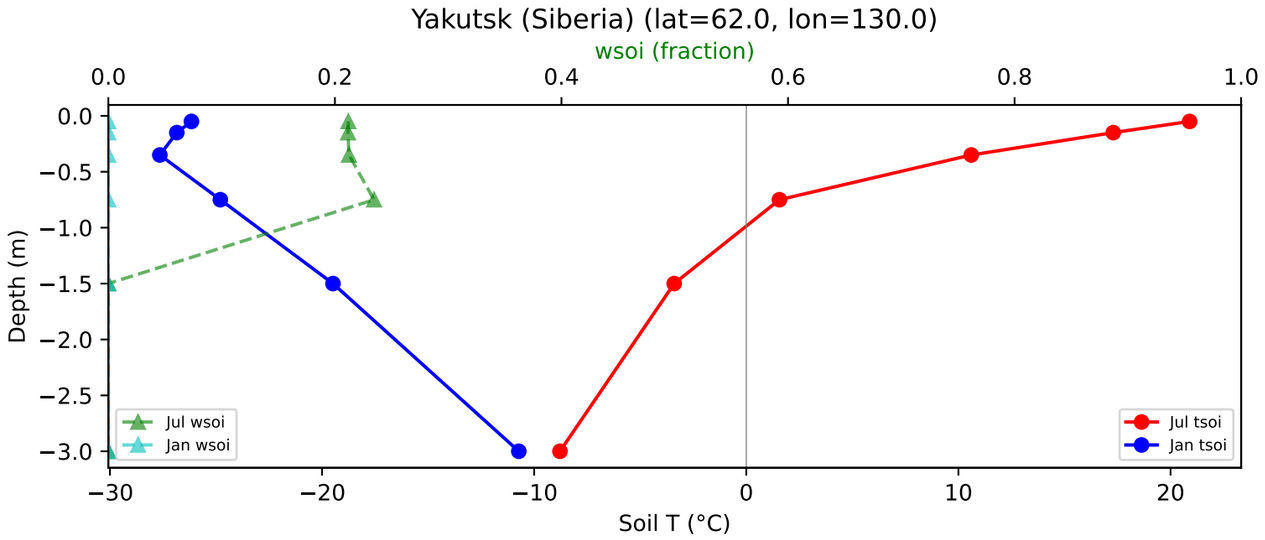
Y200 Jul drainage (mm/day)

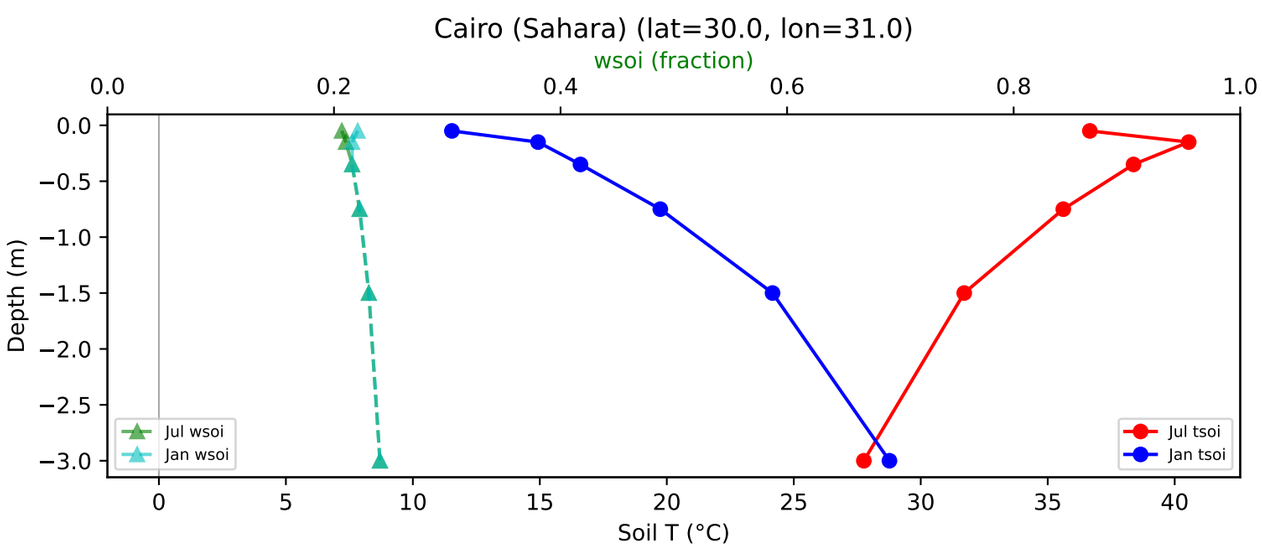


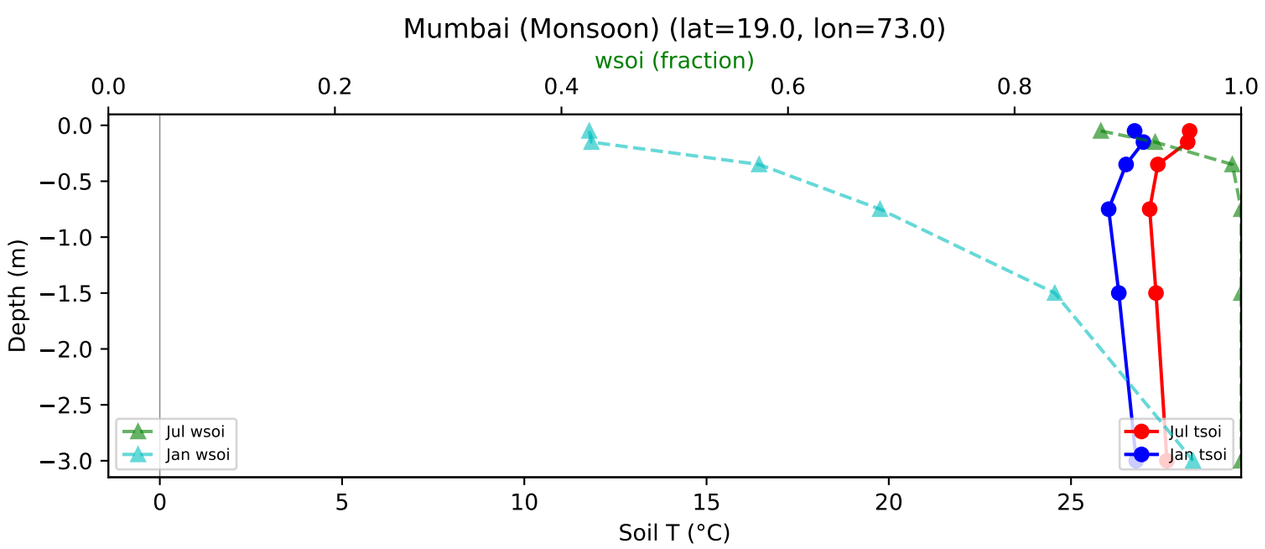
Y200 Jul ground T (°C)

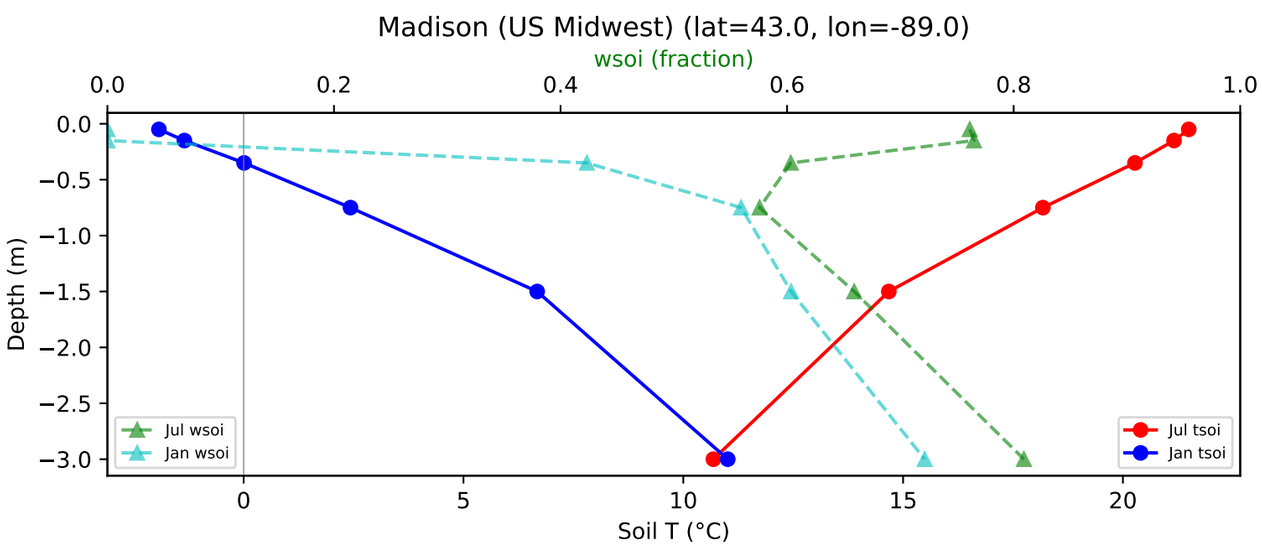


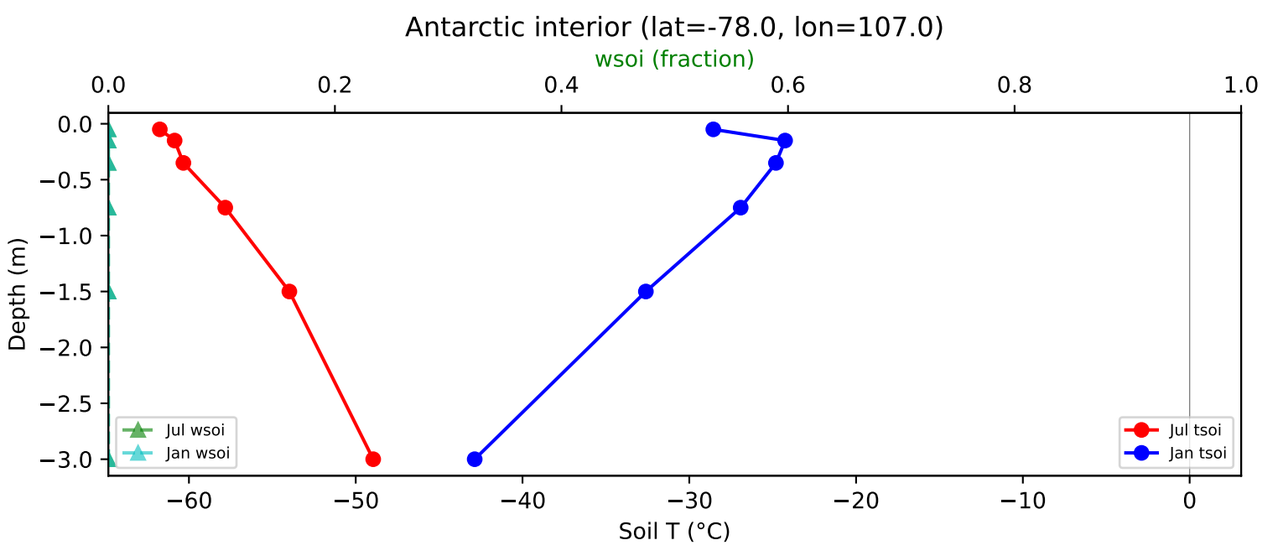

Manaus (Amazon) (lat=-3.0, lon=-60.0)
wsoi (fraction)
Depth (m)
Soil T (°C)
Jul wsoi
Jan wsoi
Jul tsoi
Jan tsoi

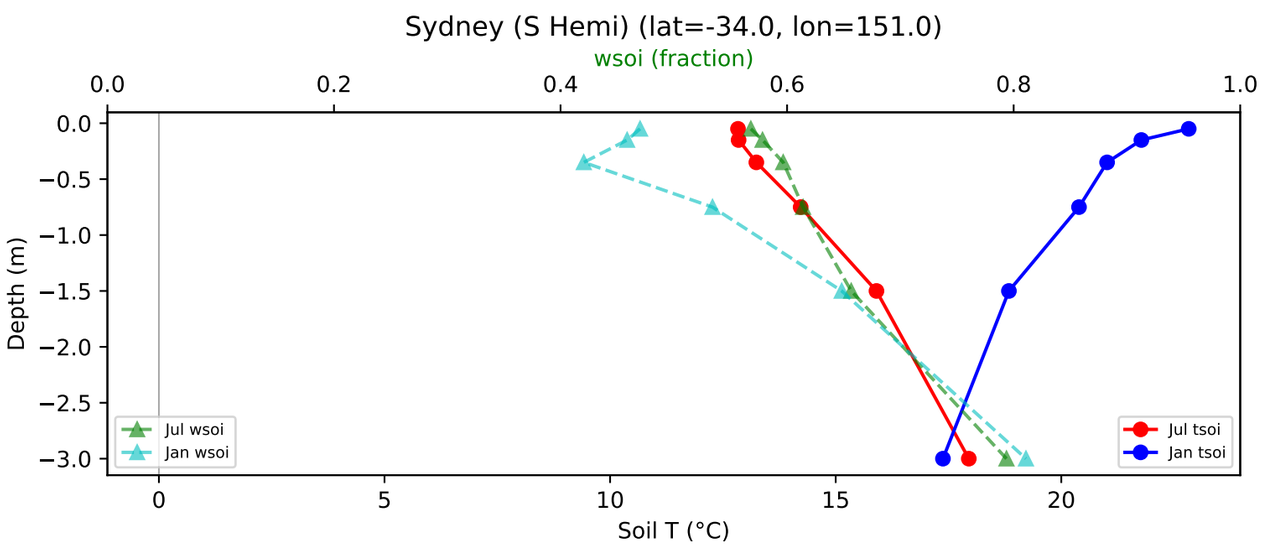

Yakutsk (Siberia) (lat=62.0, lon=130.0)
wsoi (fraction)
Depth (m)
Soil T (°C)
Jul wsoi
Jan wsoi
Jul tsoi
Jan tsoi

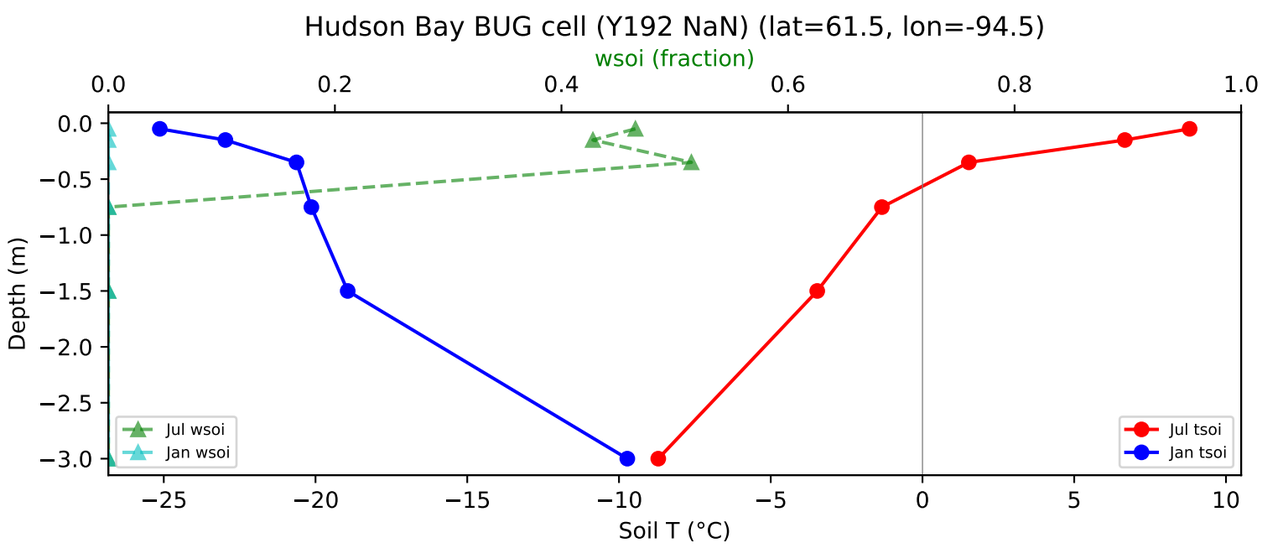

Cairo (Sahara) (lat=30.0, lon=31.0)
wsoi (fraction)
Depth (m)
Soil T (°C)
Jul wsoi
Jan wsoi
Jul tsoi
Jan tsoi


Mumbai (Monsoon) (lat=19.0, lon=73.0)
wsoi (fraction)
Depth (m)
Soil T (°C)
Jul wsoi
Jan wsoi
Jul tsoi
Jan tsoi


Madison (US Midwest) (lat=43.0, lon=-89.0)
wsoi (fraction)
Depth (m)
Soil T (°C)
Jul wsoi
Jan wsoi
Jul tsoi
Jan tsoi


Antarctic interior (lat=-78.0, lon=107.0)
wsoi (fraction)
Depth (m)
Soil T (°C)
Jul wsoi
Jan wsoi
Jul tsoi
Jan tsoi


Sydney (S Hemi) (lat=-34.0, lon=151.0)
wsoi (fraction)
Depth (m)
Soil T (°C)
Jul wsoi
Jan wsoi
Jul tsoi
Jan tsoi


Hudson Bay BUG cell (Y192 NaN) (lat=61.5, lon=-94.5)
wsoi (fraction)
Depth (m)
Soil T (°C)
Jul wsoi
Jan wsoi
Jul tsoi
Jan tsoi